MULTIMODAL FAKE DISASTER NEWS DETECTION ON TWITTER USING DEEP LEARNING

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In

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CERTIFICATE

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**Multimodal Fake Disaster News Detection on Twitter Using Deep Learning**
2	**Student Names & Numbers**	Shaik Nihajasmine (228W1A12I9) Chinni Bhanu Prakash (228W1A12E5)
3	**Name of the Guide**	Dr T. Anuradha
4	**Research Group**	Machine Learning
5	**Application Area**	Social Media
6	**Aim of the Project**	To build a multimodal system that uses text and image analysis to find disaster-related posts on Twitter and classify them into real or fake categories.
7	**Project Outcomes**	• A functional multimodal system capable of classifying disaster-related tweets as Fake or Real. • Individually fine-tuned text and image models capable of extracting meaningful semantic and visual features relevant to disaster misinformation. • A unified prediction mechanism using decision-level fusion, with evaluated performance comparison against unimodal (text-only and image-only) models.

Student Signatures:

1.
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Signature of the Guide

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TABLE OF CONTENTS

ACKNOWLEDGEMENT iii

LIST OF FIGURES v

LIST OF TABLES vi

ABSTRACT vii

CHAPTER-1 INTRODUCTION 1

1.1 Origin of the Problem 1

1.2 Basic definitions and Background 1

1.3 Problem Statement with Objectives and Outcomes 2

1.4 Related Applications 3

CHAPTER-2 REVIEW OF LITERATURE 5

2.1 Description of Existing Systems 5

2.2 Summary of Literature Study 8

2.3 Software Requirement Specification 11

CHAPTER-3 PROPOSED METHOD 15

3.1 Design Methodology 15

3.2 System Architecture 16

3.3 Description of Algorithms 18

3.4 Description of datasets, Requirements and Tools 30

CHAPTER-4 RESULTS AND OBSERVATIONS 33

4.1 Stepwise Description of Results 33

4.2 Result Analysis 55

4.3 Observations from the Work 59

CHAPTER-5 CONCLUSION 61

5.1 Conclusion 61

5.2 Future Study 61

REFERENCES

APPENDIX

LIST OF FIGURES

Figure Name	Page no
Figure 1: Proposed Multimodal System Architecture for Fake Disaster News Detection	16
Figure 2: DistilBERT Confusion Matrix	34
Figure 3: RoBERTa Confusion matrix	35
Figure 4: Confusion Matrix of ResNet50	36
Figure 5: Fusion Confusion Matrix	37
Figure 6: Training and validation loss curve for Disaster/Not	38
Figure 7: Training and validation loss curve for real/fake classification	39
Figure 8: Real/Fake Confusion Matrix (RoBERTa)	39
Figure 9: Disaster/Not Confusion Matrix (RoBERTa)	40
Figure 10: Training and validation loss curve for ResNet50 model	41
Figure 11: Final Fusion Confusion Matrix	42
Figure 12(a): Sample real disaster-related tweet describing a boat accident	43
Figure 12(b): System output for real disaster tweet showing classification results	43
Figure 13(a): Sample social media post containing misleading disaster-related claims	44
Figure 13(b): System output for fake content showing classification as non-disaster	45
Figure 14: A Snapshot of the Disaster Tweet Dashboard	47
Figure 15: Interface snapshot showing real time tweets and Heatmap	51

LIST OF TABLES

Table Name	Page no
Table 1: Summary of Literature Study	9
Table 2: Comparison of Performance metrics of different models used	58

ABSTRACT

The social media is growing fast and it has become a big source of information when something bad happens but it also helps to spread false information. This paper is about a system that uses deep learning to find out if the information about a disaster is real or not by looking at the text and the pictures. The system uses a model called RoBERTa to check if the information is about a disaster and if it is real and it also uses another model called ResNet50 to find out what kind of disaster it is from the pictures. The system then combines the results from the text and the pictures to make a reliable decision. It also tries to find out where the disaster happened and how urgent it is by looking at the words used and the location. The system can even show the disaster on a map in time to help people make decisions. The tests show that this system is better than systems that only look at the text or the pictures and it can handle the noisy information on social media very well. The social media system is very useful for keeping an eye on disasters and stopping the spread of information, about disasters.

Keywords: Decision-Level Fusion, Deep Learning, Disaster Detection, Fake News Detection, Multimodal Learning, Multitask Learning, ResNet50, RoBERTa, Social Media Analysis

CHAPTER-1

Introduction

1.1 Origin of the Problem

The rapid growth of social media platforms has transformed how the information is generated, consumed and spread. Especially during disaster events such as Earthquakes, floods, wildfires, etc., these social media platforms act as a primary source of information. As Twitter is open to post, many people post updates there, which enables the transformation of information in real time, but at the same time, there is a high chance of spread of misinformation or fake news, which could create panic in public [1].

One of the limitations is that misinformation spreads more quickly than verified information due to the emotional connection, urgency, and incorrect verification. These false reports about the disasters, rescue operations or deaths of people can cause panic in public, which reduces their trust in authorities, which prone to making bad decisions [2].

Most of the existing systems rely on textual analysis, but in real-world scenarios, most of the tweets come with both image and text, so utilising both image and text would give a better idea of the scenario.

Hence, we need an intelligent system that has the ability to identify disaster-related content and identify whether it is real or fake as well. This project addresses these drawbacks by building a deep learning-based multimodal framework for detecting disaster related-fake news [3], [4].

1.2 Basic definitions and Background

The following are the definitions required for understanding the proposed system:

Fake news: In a disaster context, fake news is the false or misleading information that is shared intentionally or unintentionally during crises. It often includes fake information and rumours which cause panic in the public [5].

Disaster-related content: Disaster-related content is the information generated during disasters, which are either man-made or natural, such as floods, earthquakes,

accidents, etc. Filtering such information from the large amount of data is necessary for identifying disaster information [6].

Multimodal Learning: Multimodal learning involves combining two or more data types such as text, image, etc., to improve the model performance. It helps to capture more information when compared to unimodal.

Deep learning: It is a subset of machine learning that uses neural networks with multiple layers to learn complex patterns. This is widely used in computer vision tasks and natural language processing.

RoBERTa (Robustly Optimized BERT Approach): RoBERTa is a high-performance transformer-based language model developed by Facebook AI. This is an improvement upon BERT by training on large datasets, using dynamic masking [7]. In this project, Roberta is fine-tuned on two classification heads to classify whether a tweet is related to:

- Disaster or not
- Real or fake

ResNet50: ResNet50 is a deep convolutional neural network (CNN) that is used for image recognition and feature extraction from visual data. In this work, it is used to identify v data and estimate disaster relevance from the image.

Decision level fusion: A fusion technique in which outputs from multiple models are combined at the final stage. In this project, a weighted fusion strategy is used, giving higher importance to textual predictions and comparatively lower importance to image predictions [8].

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement

Existing fake news detection systems mostly rely on tweet information and do not utilise visual information, which does not capture multimodal characteristics of Twitter data. Moreover, some systems are not specifically designed to identify disaster-related fake information; they are used for general categorisation of fake news [9].

Therefore, a system is required which can:

- Identify disaster-related content on Twitter
- Which filters out the fake information
- Utilises both visual and textual data for improved classification

1.3.2 Objectives

The primary objectives of this project are:

- To build a multimodal framework that filters out fake disaster tweets using textual information.
- To fine-tune RoBERTa for multitasking with two classification heads, to identify whether a tweet is related to a disaster or not and real or fake, and fine-tune ResNet50 to identify the type of disaster
- To implement decision-level fusion to combine both the outputs of textual and visual information for final classification

1.3.3 Expected Outcomes

The proposed system is expected to have these outcomes:

- A multimodal system that is capable of identifying fake disaster tweets.
- Fine-tuned models that extract textual and visual information relevant to real disasters and the type of disaster
- A decision-level fusion mechanism that combines the outputs and is evaluated against individual modals

1.4 Related Applications

The suggested system can be implemented to a number of real-life situations:

- **Disaster Management Systems:** Government agencies and emergency services may use the system to sieve credible information and improve on the response mechanisms during the occurrence of a disaster.
- **Social Media Monitoring:** such models can be integrated into social media platforms, which will automatically detect and mark suspicious disaster-related information.

- **News Verification Platforms:** The system can help journalists and fact-checking organizations to confirm the validity of information related to disasters.
- **Public Safety and Awareness:** Timely and precise dissemination of information may decrease panic and make individuals make informed choices in case of emergencies.
- **Research and Data Analysis:** The system will be able to assist in research and analysis of trends in misinformation and help to create more sophisticated detection methods [10].

CHAPTER-2

Review of Literature

2.1 Description of Existing Systems

1. **Paper Title:** Emotions-Based Disaster Tweets Classification: Real or Fake

Authors: M Alfonse, M Gawich

Year: 2023

Journal: WSEAS Transactions on Information Science and Applications

Description:

This paper proposes a classification scheme of determining whether the tweets about disaster are genuine or fake using both textual and emotional attributes. The authors suggest a mixed method in which traditional natural language processing methods are complemented by a method of emotion analysis to improve the classification. The system identifies emotion-related features (fear, sadness, anger, and trust) in the content of tweets and incorporates them into the feature set to be classified.

A labeled dataset of disaster tweets, which is sourced on Kaggle, is used to evaluate this model. The results of the experiments prove that the presence of emotional features enhances classification accuracy (88.34 to 89.39). The paper also points to emotional backgrounds as a key factor in determining the presence of misleading information related to disasters because fake news tends to capitalize on emotional appeal to gain more attention and go viral. Nonetheless, the methodology is restricted to text-based data and fails to include visual data, which is an essential part of contemporary social media content [11].

2. **Paper Title:** Fake News Detection With Credibility Scoring on Turkish Twitter Data

Authors: ÖA Fidan, E Öztürk, H Kantarci

Year: 2025

Journal: IEEE Access

Description:

This study presents a deep learning-based fake news detection framework that combines credibility scoring mechanisms based on a variety of dimensions of social media data. The system calculates credibility scores based on three significant factors, namely, text-based features, social interaction features (including likes, retweets, and replies), and user-based features (including follower count, account age, and is verified). A combination of these features is made with the help of a weighted scoring formula to get a total credibility score of each tweet.

The experiment uses big language models (LLaMA-2 and Phi-3) fine-tuned under various classification tasks (binary and multi-class credibility prediction). The data is Turkish twitter data that is coded in four levels of credibility, which are: Not Credible, Low Credible, Credible, and High Credible.

According to the results of the experiments, the use of credibility scores considerably enhances the performance, as opposed to text-only models. The system has F1-scores of 0.74 to 0.99 under various situations. The research also shows that in some situations, manual feature weighting outperforms the automatically learned weights, indicating the significance of domain knowledge in credibility assessment [12].

3. **Paper Title:** A BERT-Based Multimodal Framework for Enhanced Fake News Detection Using Text and Image Data Fusion

Authors: M Al-alshaqi, DB Rawat, C Liu

Year: 2025

Journal: Computers

Description:

The current paper presents a multimodal fake news detection system, involving a combination of both textual and visual data to enhance the classification efficiency. The suggested system involves a BERT-based model to identify contextual features of text information, and convolutional neural networks to identify visual features of related pictures. A combination of these features is then achieved by a fusion mechanism in order to achieve final classification.

The research proves that multimodal strategies are far much superior to unimodal systems and especially in situations where the false stories are backed by misleading images. The model is able to give a deeper insight into social media contents by utilizing both the semantic data of text and visual data of images.

Nevertheless, the convergence of modalities and integration of these modalities is still a difficult task particularly when text and visual data are incompatible or loosely coupled [13].

4. **Paper Title:** A Sentiment-Aware Contextual Model for Real-Time Disaster Prediction Using Twitter Data

Authors: G Song, D Huang

Year: 2021

Journal: Future Internet

Description:

The paper at hand deals with contextual modeling of real-time prediction of disasters based on Twitter inputs with the inclusion of sentiment-aware contextual modeling. The proposed method examines textual data and sentiment data to determine the occurrence of disaster-related events and predicts their events. The system underlines the significance of contextual knowledge and emotional clues of disaster patterns of social media feeds.

The combination of sentiment analysis and contextual features will allow the model to boost the detection of information related to disasters and ultimately, increase the accuracy of predictions. The approach is particularly useful for early detection and monitoring of disaster events. Nevertheless, the system is more focused on disaster prediction as opposed to fake news detection and does not specifically mention misinformation classification [14].

5. **Paper Title:** ContCommRTD: A Distributed Content-Based Misinformation-Aware Community Detection System for Real-Time Disaster Reporting

Authors: ES Apostol, CO Truică, A Paschke

Year: 2024

Journal: IEEE Transactions on Knowledge and Data Engineering

Description:

The paper outlines a distributed misinformation detection in real-time disaster reporting through the community structures of the social networks. The strategy aims at finding groups of users and content that help spread misinformation. Through content-based filtering and community detection methods, the system will be able to detect trends in the spreading of misinformation.

The suggested system is scalable and real-time processing oriented, which is why it can be used in large-scale social media scenarios. It contains information about the spread of misinformation within communities and the significance of analyzing networks at the network level. Nonetheless, the method does not explicitly utilize multimodal classification methods which are based on deep learning, and it restricts its usefulness in examining content-level features like text semantics and visual information [15].

Based on the studies above, it is clear that current methods emphasize on individual elements like emotion-based classification, credibility scoring, or community-level analysis. Nevertheless, little focus has been on combining both textual and visual modalities and authenticity detection in disaster situations. This indicates that a multimodal system is required that is able to deal with disaster relevance and misinformation detection.

2.2 Summary of Literature Study

To gain knowledge of the existing solutions, methodologies, and constraints of the field of disaster-related fake news detection and social media analysis, a thorough analysis of existing literature has been performed. The studies reviewed concern many different aspects, including the classification of texts, sentiment analysis, credibility score, multimodal learning, and community-based misinformation detection.

Based on the analysis, it can be noted that the majority of the systems available do not process multiple data modalities in a holistic manner but instead rely on textual features or concentrate on a particular aspect of the system, sentiment or credibility. Also, although there are studies that use advanced deep learning models, issues like

effective multimodal data fusion, cross-dataset generalization, and real-time applicability are not properly tackled.

Table 1 provides summarized comparative analysis of the literature reviewed, including the major details of the datasets used, algorithms or technologies used, goals of each of the studies and their strengths and limitations. The given comparative analysis contributes to pinpointing the gaps in research and inspires the necessity of a stronger multimodal approach to detecting fake disaster news.

Table 1: Summary of Literature Study

S.No	Title & Publication Details	Datasets Used	Algorithms / Technologies Used	Purpose	Remarks
1	**Title:** Emotions-Based Disaster Tweets Classification: Real or Fake [11] **Authors**: M Alfonse, M Gawich **Journal**: WSEAS **Year**: 2023	Kaggle disaster tweets dataset with labeled real/fake classes	NLP preprocessing, Machine Learning classifiers, Emotion feature extraction	To classify disaster-related tweets as real or fake using both textual and emotional features	• Improves accuracy using emotion features. • Limited to text-only data. • No multimodal or visual analysis
2	**Title**: Fake News Detection With Credibility Scoring on Turkish Twitter Data [12] **Authors**: ÖA Fidan, E Öztürk, H Kantarci **Journal**: IEEE Access **Year**: 2025	Turkish Twitter dataset (~5169 tweets) with credibility labels (NC, LC, C, HC)	LLaMA-2, Phi-3, Credibility scoring, Feature weighting, SHAP analysis	To detect fake news using credibility-based scoring mechanisms	• Achieves high performance using credibility scoring • Requires additional user/social metadata • High computational

					cost and limited generalization
3	**Title**: A BERT-Based Multimodal Framework for Enhanced Fake News Detection Using Text and Image Data Fusion [13] **Authors**: M Al-alshaqi, DB Rawat, C Liu **Year**: 2025 **Journal**: Computers	Multimodal dataset (text-image pairs)	BERT, CNN, Multimodal fusion techniques	To improve fake news detection using combined text and image data	• Shows effectiveness of multimodal learning • Difficulty in aligning text and image features • Performance depends on data quality and fusion strategy
4	**Title**: A Sentiment-Aware Contextual Model for Real-Time Disaster Prediction Using Twitter Data [14] **Authors**: G Song, D Huang **Year**: 2021 **Journal**: Future Internet	Twitter dataset	Sentiment analysis, Contextual modeling	To predict disaster events using sentiment-aware approaches	• Uses sentiment to improve detection • Does not perform fake news classification • Lacks multimodal integration
5	**Title**: ContCommRTD: A Distributed Content-Based Misinformation-	Social network data	Distributed systems, Community detection algorithms	To detect misinformation spread in real-time disaster scenarios	• Scalable for real-time systems • Does not analyze content

	Aware Community Detection System for Real-Time Disaster Reporting [15] **Authors**: ES Apostol, CO Truică, A Paschke **Year**: 2024 **Journal**: IEEE Transactions on Knowledge and Data Engineering				semantics deeply • No multimodal or deep learning-based classification

2.3 Software Requirement Specification

The Software Requirement Specification (SRS) is the statement of the functional and non-functional requirements of the proposed system named Multimodal Fake Disaster News Detection on Social Media Using Deep Learning. This system is a full-stack application, which combines deep learning models and an interactive dashboard to monitor and analyze in real-time [3].

2.3.1 System Overview

The suggested system works with the input of both textual and visual information to process social media data and reveal disaster-related information and misinformation. The system is based on a multimodal pipeline, which comprises:

- **Frontend:** React based user interaction and visualization interface.
- **Backend**: Flask-based server to process API requests and model inference.
- **Database**: MySQL to store the disaster-related data that is validated.

To perform the textual component, a fine-tuned RoBERTa model with multitask learning is used:

- Disaster vs Non-Disaster classification.

- Real vs Fake classification.

The visual part applies ResNet50 to recognize the type of disasters based on images. The results are then fused together with a decision level fusion mechanism to enhance the reliability of the prediction [16].

2.3.2 Functional Requirements

The system shall:

- Take input as text(tweet) and image
- Fetch real-time tweets using Twitter API
- Preprocess text and image data before passing to models
- Perform multitask classification using RoBERTa:
    - Disaster or Non-disaster
    - Real or Fake
- Identify disaster type using ResNet50
- Extract location information from tweet using Named Entity Recognition (NER)
- Convert locations into geographical coordinates using geocoding
- Estimate urgency suing Sentiment analysis
- Apply decision-level fusion to combine the predictions
- Store only valid inputs
- Display results in interactive dashboard
    - real-time feed
    - geospatial visualization [17]

2.3.3 Non-Functional Requirements

Performance:

- The system must provide predictions in a couple of seconds.
- Effective management of streaming data.

Scalability:

- Capacity to deal with growing data volume and API requests.

Usability:

- Easy to use dashboard by non-technical user.

Reliability:

- Stable and predictable model predictions.

Maintainability:

- Modular design to enable models and components to be updated easily.

2.3.4 Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 10 GB
- GPU: Required for training and faster inference

2.3.5 Software Requirements

- Operating System: Windows
- Programming Language: Python
- IDE: Jupyter Notebook
- **Libraries and Frameworks:**
  - PyTorch
  - Transformers (Hugging Face)
  - NumPy, Pandas
  - Scikit-learn
  - OpenCV
  - Flask
  - React
  - Leaflet (for map visualization)

2.3.6 System Constraints

- Limited availability of labeled multimodal datasets

- High computational cost of deep learning models
- Dependency on pre-trained models
- Variability and noise in social media data

2.3.7 Assumptions

- Input data is relevant and processable
- Pre-trained models are accessible
- Required computational resources are available
- Internet access is available for API-based data collection [18]

CHAPTER-3

Proposed Method

3.1 Design Methodology

The suggested system embraces a multimodal deep learning approach to identify fake news on disasters by combining both text and image content. Multimodal systems have been demonstrated to perform better than unimodal systems with the implementation of complementaryness of the various sources of data.

The general architecture is made of four key steps: data preprocessing, model training, multimodal fusion and final classification. First, text and image data is handled independently to be compatible with their models. Preprocessing operations on text data include cleaning, normalization, and tokenization, which play a crucial role in enhancing transformer-based models. On the same note, image information is downscaled and brought to the normalized range of convolutional neural networks input.

To do textual analysis, a fine-tuned RoBERTa model is trained using a multitask learning strategy. RoBERTa and other transformer-based architectures are very useful in extracting contextual relationships in textual information using self-attention mechanisms, and can be applied to detect fake news and classify disasters. The multitask system allows the model to learn the disaster relevancy and authenticity at the same time enhancing efficiency and sharing of features.

To analyze the images visually, a ResNet50 model is used to derive hierarchical features of images. The application of residual networks in computer vision has become popular because it enables the training of deep architures through skip connections, reducing the vanishing gradient problem. The model is optimized to categorize the types of disasters, so meaningful visual patterns can be extracted.

A decision-level fusion strategy is adopted to combine the outputs of the two modalities. Here, the text and image models are used to make predictions which are averaged together through weighted averaging. Decision-level fusion is computationally inexpensive and enables the independent maximization of every modality, which is appropriate to multimodal systems. This work puts more weight on

textual predictions because they are more semantically relevant in detecting fake news.

The last system is used to generate disaster classification, fake news detection and disaster type prediction. The modular design guarantees scalability and flexibility, with the ability to expand the system to real-time applications and other modalities to be added.

3.2 System Architecture

Figure 1 shows the system architecture of the proposed multimodal fake news detector of disaster news. The design is such that the textual and visual data are processed concurrently and their outputs are fused through a decision-level fusion mechanism. Such a modular structure allows making good use of semantic and visual information to enhance classification.

The general system is made up of three fundamental parts, which include the text processing pipeline, image processing pipeline and the fusion and output module. Such multimodal architectures are actively used to use complementary data content in various data sources to enhance robustness and accuracy.

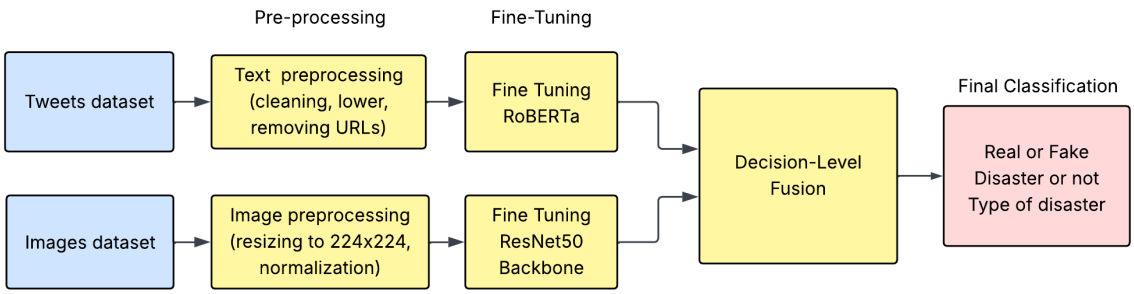

Figure 1: Proposed Multimodal System Architecture for Fake Disaster News Detection

1. Text Processing Pipeline

The text pipeline starts with the tweet data being fed into it and the data is processed with remedial steps including cleaning, lowercasing, and deleting of URLs. The steps are useful in the minimization of noise and the enhancement of model performance.

A fine-tuned RoBERTa model is then applied to the already preprocessed text. The model produces two outputs and contextual representations of the input text:

- Disaster probability
- Real/Fake probability

These outputs reflect the topicality of the content to the disaster events as well as its authenticity.

2. Image Processing Pipeline

The pipeline of images works with images related to disasters that are linked to social media posts. The images are initially processed by resizing the images to 224 x 224 pixels and normalising the pixel values to the input specifications of the convolutional neural networks.

The processed images are further sent through a fine-tuned ResNet50 model that retrieves the hierarchical visual features and determines the disaster type. The output of this model is then transformed into a disaster probability score so as to be compatible with the fusion module.

ResNet-based architectures have been popular in image classification tasks since they are capable of learning deep feature representations.

3. Decision-Level Fusion Module

The text and image pipelines are fused at the decision-level through a fusion method. This approach uses both modalities to make predictions at the last stage instead of combining features at earlier stages.

A weighted fusion approach is used, with textual predictions being given a higher weight than image-based predictions. This is due to the fact that textual data normally has a more powerful semantic meaning in detecting fake news.

The fusion process makes the system more equipped to deal with ambiguous or incomplete data by exploiting the complementary capabilities of both modalities.

4. Final Classification Module

The final module produces the overall system output, which includes:

- Real or Fake classification

- Disaster or Non-Disaster classification
- Type of disaster

These outputs give a holistic perception of the input data and can help identify misinformation effectively in case of a disaster.

The architecture is to be scalable and modular with the possibility of adding any other component in the future like real-time data streams or more advanced fusion techniques.

3.3 Description of Algorithms

In this section, the basic algorithms used to develop the system are described. It developed the text-based classification model into a multimodal deep learning model. Transformer-based models were initially applied to analyze text, and further developed to include convolutional neural networks to analyze images. The last system combines both modalities with the high-level deep learning approaches.

3.3.1 DistilBERT (Pretrained Transformer Model)

DistilBERT is a lightweight implementation of BERT that preserves its performance to the greatest extent and minimizes the computational complexity. It is trained via knowledge distillation, in which a smaller student model is trained to replicate a larger teacher model.

The training goal is a combination of language modeling loss and distillation loss.

Algorithm 1: DistilBERT Algorithm

Require: Tokenized input sequence

$$X = (x_1, x_2, \dots, x_n)$$

Ensure:

Contextual representations $\boldsymbol{H}$

Step 1: Input Embedding

Convert tokens into embeddings:

$$e_i = Embedding(x_i), \quad E = (e_1, e_2, \dots, e_n)$$

Step 2: Masked Language Modeling Setup

Define masked index ***M*** set:

$$M \subset \{1,2,\dots,n\}$$

Replace selected tokens with [MASK] to obtain $\tilde{\boldsymbol{X}}$

Step 3: Transformer Encoding

Pass embeddings through ***L*** transformer layers:

$$H^{(l)} = Transformer\left(H^{(l-1)}\right), \quad l = 1,\dots,L$$

Final output:

$$H = H^{(L)}$$

Step 4: Self-Attention Computation

Compute attention as:

$$Attention\ (Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Step 5: Masked Language Modeling Loss

Compute loss only for masked tokens:

$$\mathcal{L}_{MLM} = -\frac{1}{|M|}\sum_{i\in M}\log P\left(x_i|\tilde{X}\right)$$

Step 6: Distillation Loss

The student model learns from the teacher model using KL divergence:

$$\mathcal{L}_{distill} = \sum_{i\in M}\sum_{v\in V} P_{teacher}(v|\tilde{X},i)\ \ log\frac{P_{teacher}(v|\tilde{X},i)}{P_{student}(v|\tilde{X},i)}$$

Step 7: Total loss Function

$$\mathcal{L} = \alpha\mathcal{L}_{MLM} + \beta\mathcal{L}_{distill}$$

Step 8: Parameter update

Update model parameters using gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

Step 9: Output

Return contextual representations:

$$H = (h_1, h_2, \dots, h_n)$$

Terms Explanation:

- $X = (x_1, x_2, \dots, x_n)$

Input sequence tokenized, with each x_i being a token of a word or subword in the input text.

- $Embedding(e_i)$

Each token x_i is converted into a dense vector representation e using an embedding layer, forming $E = (e_1, e_2, \dots, e_n)$. These embedding represent semantic meaning of words.

- Masked Set $\boldsymbol{M}$

A random sample of token positions to be masked. Tokens at these positions are replaced with a special [MASK] token to train the model in predicting missing words.

- $\tilde{X}$ (Masked Input)

The altered input sequence in which the chosen tokens are substituted with [MASK], which is applied in masked language modeling.

- $H^{(l)}$

Representations at layer *l* of the transformer. Every layer narrows down the contextual interpretation of the input sequence.

- $Transformer()$

A stack of encoder layers with self-attention and feed-forward networks, which are used to learn relationships between tokens.

- Q, K, V $(Query, Key, Value)$

Vectors derived from embeddings used in self-attention:

Query = current token

Key = all tokens to compare

Value = actual information of tokens

- d_k

Scale of key vectors, which are used to scale the attention scores to ensure numerical stability.

- Attention mechanism

Computes the importance of each token with respect to others and generates context-aware representations.

- $\mathcal{L}_{MLM}$ (Masked Language Modeling Loss)

Loss computed on masked tokens, which is the degree to which the model predicts the original tokens.

- $\mathcal{L}_{distill}$ (Distillation loss)

Compares student and teacher model predictions based on KL divergence, which assists the student model in learning effectively.

- $P_{teacher}, P_{student}$

Probability distributions over vocabulary predicted by teacher and student models respectively.

- α, β

Weights to balance masked language modeling loss and distillation loss.

- θ

Parameters (weights) of the model that are updated during training.

- η

Step size controlled by learning when updating the parameters.

- $H = (h_1, h_2, \dots, h_n)$

Final contextual representations of all tokens, including relationships and meaning in the sequence.

3.3.2 RoBERTa (Robustly Optimized BERT Approach)

Transformer-based architectures have also greatly enhanced natural language processing tasks by exploiting long-range dependencies with self-attention mechanisms. RoBERTa is an optimized form of BERT which provides a higher training efficiency and better downstream performance.

RoBERTa is here fine-tuned with the help of a multitask learning framework. The input text is processed by a common encoder and two distinct classification heads are involved in predicting the disaster relevance and authenticity. The multitask learning allows common representation of features, enhancing generalization, and minimizing computational redundancy.

Algorithm 2: RoBERTa Algorithm

Require: Tokenized input sequence

$$X = (x_1, x_2, \dots, x_n)$$

Ensure:

Contextual representations $\boldsymbol{H}$

Step 1: Input Embedding

Convert tokens into embeddings:

$$e_i = Embedding(x_i), \quad E = (e_1, e_2, \dots, e_n)$$

Step 2: Dynamic Masking

Randomly mask tokens at runtime → X_{masked}

$$M \subset \{1,2, \dots, n\}, \;\; \widetilde{x_i} = \begin{cases} [MASK] & i \in M \\ x_i & otherwise \end{cases}$$

Step 3: Transformer Encoder

Pass through ***L*** layers:

$$H^{(0)} = E$$

$$H^{(l)} = Transformer\left(H^{(l-1)}\right), \quad l = 1, \dots, L$$

$$H = H^{(L)}$$

Step 4: Multi-Head Attention

For each head:

$$Attention^{(h)}\ (Q, K, V) = softmax\left(\frac{Q^{(h)}\left(K^{(h)}\right)^T}{\sqrt{d_k}}\right)V^{(h)}$$

Step 5: Output Projection

Concatenate heads and apply linear transformation

$$MultiHead(Q, K, V) = Concat\ (head_1, \dots, head_H)W^O$$

Step 6: Loss Function

Compute MLM loss:

$$\mathcal{L} = -\frac{1}{|M|}\sum_{i \in M} \log P\left(x_i | \tilde{X}\right)$$

Step 7: Parameter update

Apply gradient descent

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

Step 9: Output

Return contextual representations ***H***:

$$H = (h_1, h_2, \dots, h_n)$$

Terms Explanation:

- $X = (x_1, x_2, \dots, x_n)$

Input sequence that is tokenized and each x_i is a tokenized word or subword.

- $Embedding(e_i)$

Each token x_i is converted into a dense vector representation e using an embedding layer, forming $E = (e_1, e_2, \dots, e_n)$. These embedding represent semantic meaning of words.

- Masked Set $\boldsymbol{M}$

A subset of token positions selected randomly that are used to mask in training.

- Dynamic Masking

In contrast to BERT, masking is dynamically done during training, such that various tokens are masked each epoch, enhancing model generalization.

- $\tilde{X}$ (Masked Input)

Input sequence in which the tokens in set M are substituted with [MASK], and others are not.

- $H^{(l)}$

Representation at layer l of the transformer encoder hidden. Every layer narrows down contextual knowledge.

- $Transformer()$

Layers of encoders with multi-head self-attention networks and feed-forward networks, layer by layer, to learn relationships between tokens.

- Q, K, V $(Query, Key, Value)$

Vectors obtained as embeddings in attention:

Query = current token representation

Key = Tokens of comparison.

Value = forwarded information.

- Multi-Head Attention

Breaks down attention into various heads to acquire various contextual relationship at once, enhancing representation power.

- d_k

Scale measure of key vectors that scale attention scores to be numerically stable.

- Attention mechanism

Calculates the significance of each token in comparison to others and produces context-sensitive results.

- Output Projection W^O

A learnable weight matrix that transforms the outputs of many attention heads into one representation.

- $\mathcal{L}_{MLM}$ (Masked Language Modeling Loss)

Masked Language Modeling loss, which only considers masked tokens, quantifies prediction accuracy.

- θ

Parameters of the model (weights) modified in the course of training.

- η

Step size controlled by learning when updating the parameters.

- $H = (h_1, h_2, \dots, h_n)$

Final contextual images of tokens at the end of all transformer layers.

3.3.3 ResNet50 Backbone (Deep Residual Network)

ResNet50 is a deep convolutional neural network which applies the concept of residual learning to allow training of very deep networks. Skip connections enable the flow of gradients without vanishing gradient issues.

The use of deep convolutional neural networks in image classification problems is based on the fact that they have the capacity to learn hierarchical features in a

representation. Residual networks (ResNet) add skip connections to enable the effective training of deep architectures by eliminating gradient degradation problems.

The ResNet50 is fine-tuned in this system to identify the type of disaster in images. The network defines visual features using convolutional layers and residual blocks and the last classification layer classifies the type of disaster. These are then transformed into disaster probability scores to be used in conjunction with textual predictions.

Algorithm 3: ResNet50 Algorithm

Require:

Input image $\boldsymbol{X}$

Ensure:

Predicted class probabilities $\boldsymbol{P}$

Step 1: Input Processing

Resize image and normalize

$$X' = Normalize(Resize(X))$$

Step 2: Convolution Layer

Apply convolution:

$$F_1 = \sigma(W_1 * X' + b_1)$$

Step 3: Residual Block

Compute:

$$F_2 = \sigma(W_2 * F_1 + b_2)$$

$$F_3 = W_3 * F_2 + b_3$$

Step 4: Skip Connection

Add input to output:

$$F_{out} = F_1 + F_3$$

Step 5: Activation

Apply ReLU:

$$F_{out} = \max(0, F_{out})$$

Step 6: Global Average Pooling

$$f_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F_{out}^{(c)}(i,j)$$

Step 7: Fully connected layer

$$z = Wf + b$$

Step 8: Softmax

$$P_k = \frac{\exp(z_k)}{\sum_{j=1}^{C} \exp(z_j)}, \quad k = 1, \dots, C$$

Step 9: Loss function

$$\mathcal{L} = -\sum_{k=1}^{C} y_k log P_k$$

Step 10: Backpropagation

Update weights

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

Step 11: Output

Return contextual representations ***P***:

$$P = (P_1, P_2, \dots, P_C)$$

Terms Explanation:

- X

Image that is fed into the model to be classified.

- X' (Preprocessed Image)

After resizing and normalizing to fit model input requirement and enhance training stability.

- W, b

Convolutional and fully connected learnable weights and biases.

- * (Convolutional Operation)

Operation that uses filters on the image to extract spatial features like edges, textures and patterns.

- $\sigma\ (Activation\ function)$

Non-linear (e.g., ReLU) layered after convolution in order to add non-linearity.

- F_1, F_2, F_3

Maps of intermediate features produced in various layers of the network modeling learned visual features.

- Residual Block

One of the main components of ResNet that enables the network to learn the mappings as residual mappings rather than direct mappings, which enhances training of deep networks.

- Skip connection

Direct connection that adds input F_1 to output F_3:

$$F_{out} = F_1 + F_3$$

Helps avoid vanishing gradient issue and enhance the model performance.

- ReLU Activation

Maximizes *max(0,x)* to make it non-linear and eliminate negative values.

- Global Average Pooling

Contrasts each feature map to a single value through spatial dimension averaging, decreasing parameters and avoiding overfitting.

- f_c

Obtained feature vector (after pooling), which depicts condensed image features.

- Fully Connected Layer

Combines the extracted features to come up with final classification scores.

- z

Before the use of softmax, raw output scores (logits).

- Softmax function

Converts a logits into a probability distribution on classes:

$$P_k = \frac{\exp(z_k)}{\sum_{j=1}^{C} \exp(z_j)},$$

- $P = (P_1, P_2, \dots, P_C)$

Last guesses about the probability of each class.

- y_k

Actual class (ground truth label).

- Loss Function (Cross-Entropy)

Makes difference between the predicted probabilities and actual labels.

- θ

Parameters of the model that are updated during training.

- η

Learning rate Controlling update step size.

The pretrained DistilBERT and RoBERTa are fine-tuned on textual classification, whereas the ResNet50 model is fine-tuned on disaster image classification. The ultimate system combines these models with decision level fusion.

3.3.4 Decision-Level Fusion

To achieve modularity and minimize complexity, multimodal systems tend to integrate the output of multiple models at the decision level. Decision-level fusion enables independent maximization of each modality and enhances robustness through the use of complementary information.

In this work, the final disaster probability is computed as:

$$P_{final} = w_1 \times P_{text} + w_2 \times P_{image}$$

where $w_1 > w_2$ to prioritize textual information.

3.4 Description of datasets, Requirements and Tools

This section explains the datasets to be trained and evaluated, hardware, software needs and tools to develop the proposed multimodal fake disaster news detecting system. The system combines text and visual sources of information to accomplish disaster relevance, authenticity and disaster type.

3.4.1 Datasets

The suggested system is based on several datasets to help with both visual and textual elements. The datasets are chosen to deal with two main tasks:

i. disaster vs. non-disaster classification.
ii. real vs fake classification.
iii. disaster type classification using images.

By integrating different datasets, data diversity is boosted, the bias is mitigated, and the overall ability of the model to generalize in the real-life situations of social media is improved.

3.4.1.1 PHEME Dataset

The real vs fake classification is done on the PHEME dataset of tweets. It contains annotated Twitter discussions about what is going on in the real world, in which every example is tagged regarding its veracity (true, false, or unverified).

In this work, the data is changed into a binary classification task (real vs fake) to correspond to the multitask learning goal of the RoBERTa model. It is a high-quality

ground truth dataset to detect misinformation, and is also employed in rumor verification tasks [19].

3.4.1.2 Disaster-related Text Datasets

Several publicly available datasets of disaster related tweets are pooled together to carry out a disaster vs non-disaster classification. Such datasets consist of tweets related to floods, earthquakes, wildfires, and accidents.

The data sets are processed and consolidated into a binary classification format with the tweets being classified as disaster or non-disaster. The head of the multitask RoBERTa model which is trained on this composite set is the disaster classification head.

It enhances the strength of the model and enhances generalization [20], [21] through the variety of language pattern, types of events, and writing styles.

3.4.1.3 Disaster Image Datasets

The ResNet50 model is trained on several disaster image datasets to classify these disasters in terms of their type. These datasets consist of images of different disaster types like floods, wildfires, earthquakes and infrastructure damage.

Preprocessing of the images is done by resizing and normalization after which they are inputted into the model. ResNet50 is trained to categorize the types of disasters, and the results will be utilized in the decision level fusion process [22], [23].

The variety of image datasets enables the model to acquire discriminative visual features and enhances the classification tasks in real-world applications [24], [25].

3.4.2 Requirements

Hardware Requirements

- **CPU**: Intel i5 or more.
- **RAM**: 8 GB (16 GB preferred) or more.
- **GPU**: needed to train and make inferences efficiently on deep learning models.
- **Storage**: 10 GB free storage or more.

Software Requirements

- **OS**: windows or Linux.

- **Programming Language:** Python
- **Development Environment:** Jupyter Notebook / VS Code.

3.4.3 Tools and Technologies

The system is applied with the following tools and frameworks:

- **PyTorch**: It is used to develop and train deep learning models (RoBERTa and ResNet50).
- **Hugging Face Transformers:** Transformers implemented to implement and fine-tune the RoBERTa model.
- **Scikit-learn:** Evaluation metrics, including accuracy, precision, recall and F1-score.
- **NumPy and Pandas:** Used for data preprocessing and manipulation
- **OpenCV:** Image preprocessing functions.
- **Flask:** API development and integration of trained models.
- **React (with Vite):** This is used to develop the interactive frontend dashboard.
- **MySQL:** Stored in MySQL to store filtered and validated disaster related data.
- **Leaflet containing the OpenStreetMap:** The leaflet will be utilized in geospatial visualization of the disaster sites on the dashboard.

CHAPTER-4

Results and Observations

4.1 Stepwise Description of Results

This part involves the step-by-step discussion of the findings of the suggested multimodal fake disaster news detection system. The analysis comprises dataset preparation, model training behavior, single model performance, fusion-based results and real-world prediction examples. These outcomes are evaluated by the use of standard evaluation methods, such as loss curves, accuracy trends, and confusion matrices, which are widely applicable to evaluate the performance of deep learning models.

4.1.1 Dataset Splitting

The data in this work is both textual and visual data that is gathered via various sources, such as rumor datasets and disaster-related tweet datasets , as well as disaster image datasets.

The entire data set was subdivided into:

- 70% Training Data
- 30% Testing Data

This division guarantees the model is trained on enough data and it still has an independent set of data to be evaluated without bias. Correct dataset splitting is a necessity to prevent overfitting and to be able to generalize deep learning models.

4.1.2 Baseline Text-Based Model

First, a text based classification model was created to determine whether a tweet is disaster related or not. In this task, the DistilBERT model was used because it can be applied to contextual relationships in text and is computationally efficient.

Tweet labeled data were used to train the model and its performance was measured using a confusion matrix.

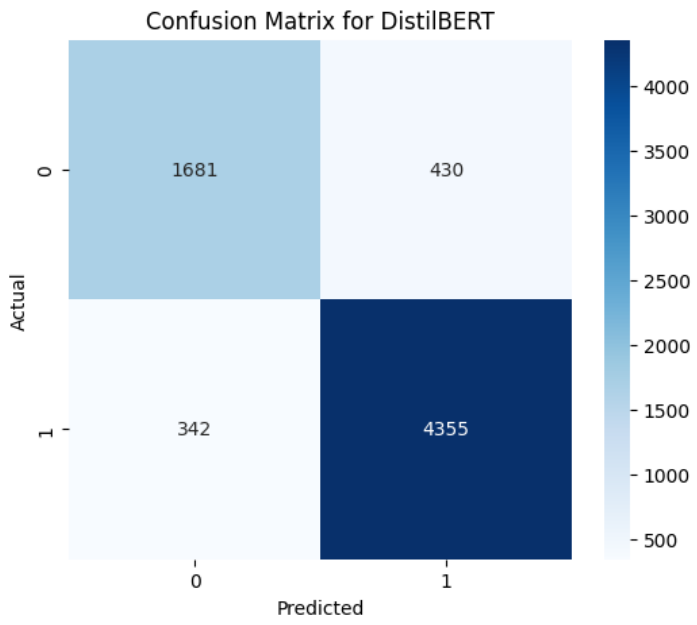

Figure 2: DistilBERT Confusion Matrix

The obtained confusion matrix of the DistilBERT model is provided in **Figure 2**. We can see that many samples are correctly classified as the values along the diagonal are high. In particular, 4355 disaster tweets and 1681 non-disaster tweets are rightly predicted.

Nevertheless, there are also a certain amount of misclassifications that are noticeable. False positives refer to the fact that about 430 non-disaster tweets are mistaken as disaster-related. On the same note, there are 342 misclassified (false negative) disaster tweets that are not classified as such.

This trend implies that the model is efficient at accumulating general contextual clues, but it does not perform well with tweets that may have ambiguous language, use indirect references, or have words similar in spellings between the scenario of disaster and non-disaster. Such cases make it difficult for the model to clearly distinguish between the two classes.

On the whole, the DistilBERT model could serve as a good starting point in the disaster classification, although the occurrence of false positives and false negatives suggests that further refinements should be made, which in turn stimulates the shift to more complex and multimodal models at the next stage.

4.1.3 Multimodal Classification Model

In order to enhance the functionality of the base system, the visual information was included along with the textual information, and a multimodal classification system was developed. The text-based classification with RoBERTa, and the image-based disaster type classification with ResNet50 were applied in this stage. The results of the two models were also fused at a decision level through a decision-level fusion method.

4.1.3.1 Text Model – RoBERTa

DistilBERT was substituted with RoBERTa in order to improve contextual knowledge of tweets. As a more optimized transformer model, RoBERTa enhances the expression of linguistic patterns that occur in the social media text.

The confusion matrix of the RoBERTa model is presented in **Figure 3**. Many of the samples are correctly classified and the correct predictability is reflected in the diagonal numbers with 5609 disaster tweets and 2397 non-disaster tweets being correctly predicted.

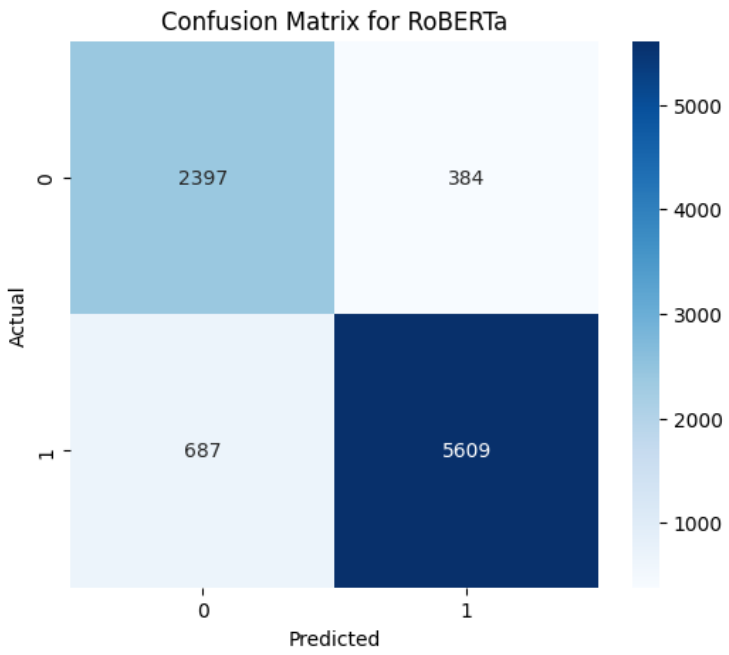

Figure 3: RoBERTa Confusion matrix

Nevertheless, there are still misclassifications. There are 384 false non-disaster tweets and 687 disaster tweets which are misclassified as non-disaster. This is a sign of better

learning capability compared to the baseline model, but there are difficulties in dealing with ambiguous or contextually complex tweets.

4.1.3.2 Image Model – ResNet50

To include visual features, ResNet50 model was employed in classifying the images under various disaster categories like cyclone, earthquake, flood, infrastructure damage, landslide, urban fire, wild fire and non-damage.

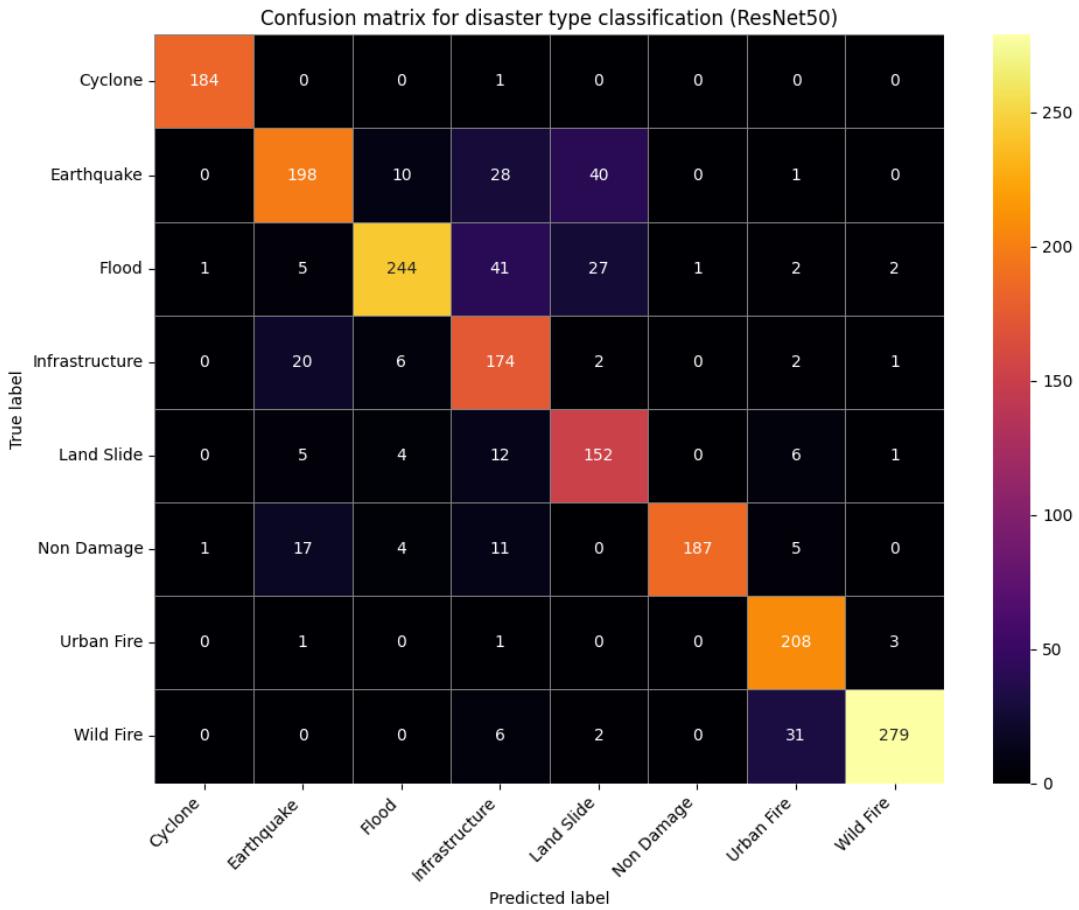

Figure 4: Confusion Matrix of ResNet50

The confusion matrix of disaster type classification is shown in **Figure 4**. The bulk of predictions is along the diagonal, meaning that the model is able to recognize most of the disaster types. An example of classes with high correct classification performance includes Cyclone (184), Flood (244), Urban Fire (208) and Wild Fire (279).

There is however some confusion between similar classes that are visually similar. To illustrate, flood images are at times incorrectly categorized as infrastructure damage or landslide and earthquake images are misclassified as flood and landslide. Such

conduct is a result of the commonalities in visual characteristics like debris, water, and structural damage, shared between several types of disasters.

4.1.3.3 Decision-Level Fusion Model

A decision-level fusion method was introduced to take advantage of the advantages of using both the textual and visual models. The approach uses a combination of both predictions of RoBERTa and ResNet50, to obtain a final classification result.

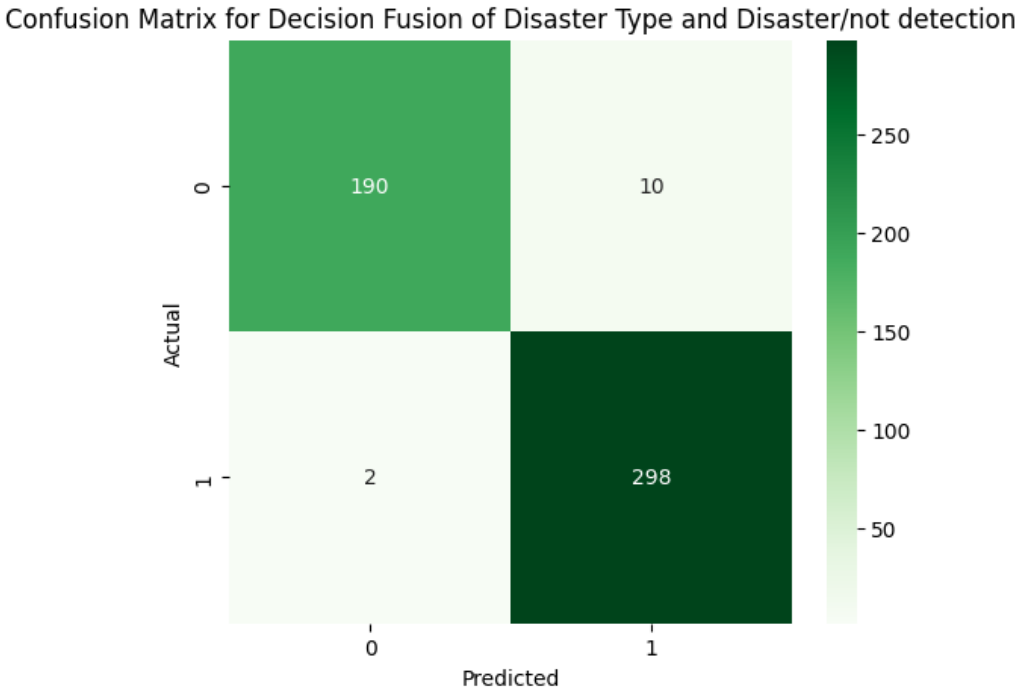

Figure 5: Fusion Confusion Matrix

The confusion matrix of the fusion model is presented in **Figure 5**. Many correct predictions are noted with 298 disaster cases and 190 non-disaster cases being correctly predicted.

There are few misclassifications (only 10 false positives and 2 false negatives). This shows that the fusion method is effective in enhancing performance in classification because it addresses ambiguities that exist in each modality.

In general, the multimodal framework helps to make the system more robust as the textual context is supplemented by visual evidence, making it a more accurate and reliable prediction than single-modality models.

4.1.4 Proposed Multitask Multimodal Model

The system proposed extends the multimodal model to multitask learning of textual data. The RoBERTa model in this approach has two classification heads to concurrently classify disasters/non-disasters and real/fake. The two tasks have a similar contextual representation that enables the model to acquire richer semantic features of the input text.

4.1.4.1 Multitask RoBERTa

The loss curves are used to demonstrate the training behavior of the multitask model.

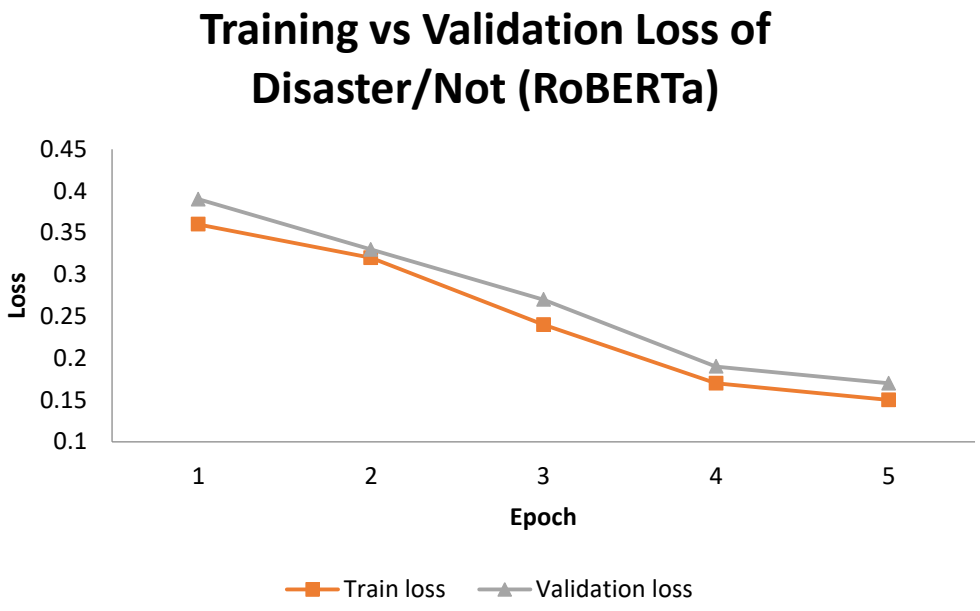

Figure 6: Training and validation loss curve for Disaster/Not

In **Figure 6**, the training and validation loss of disaster/non-disaster classification are presented. It is noted that the loss gradually drops to about 0.36 and 0.15 (training) and 0.39 and 0.17 (validation) between the epochs, which is a sign of constant learning. The slight difference between the training and valid loss indicates that the model is generalizing with no severe overfitting.

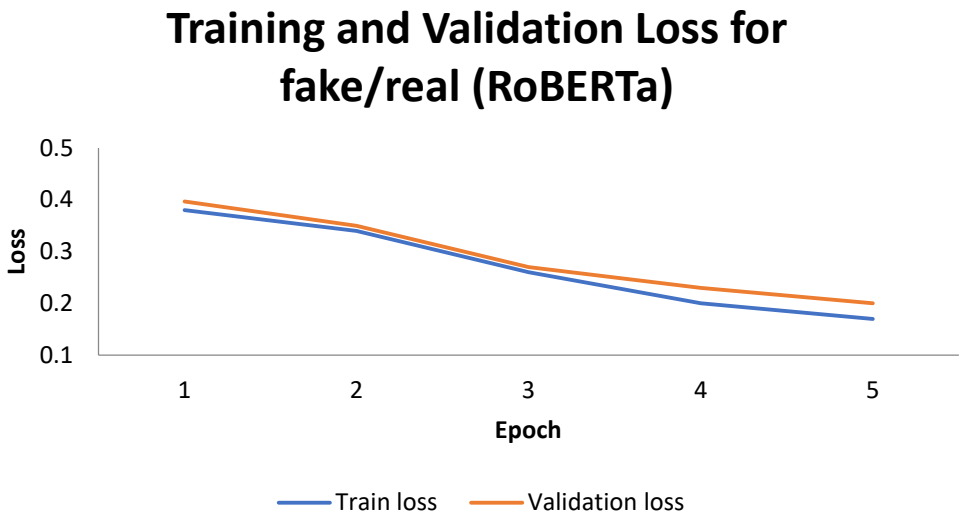

Figure 7: Training and validation loss curve for real/fake classification

The loss curves of real/fake classification are shown in **Figure 7**. The same downward trend is evident, as training and validation loss drops to approximately 0.38 and 0.40, respectively. The presence of this consistency in both tasks suggests that there is effective joint learning that involves shared representations.

4.1.4.2 Confusion Matrices for Text-Based Tasks

Confusion matrices of both tasks are used to assess the classification performance of the multitask RoBERTa model.

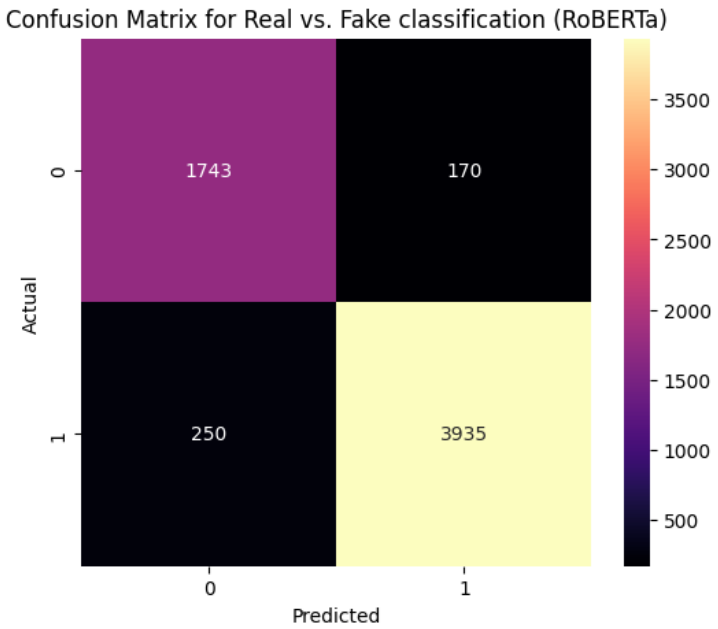

Figure 8: Real/Fake Confusion Matrix (RoBERTa)

The confusion matrix of real/fake classification is indicated in **figure 8**. The model accurately classifies 3935 real data and 1743 fake data, which shows good performance. But, 170 false positives are wrongly identified as true, and 250 true positives are false, so it is not that easy to identify obscure or deceptive content.

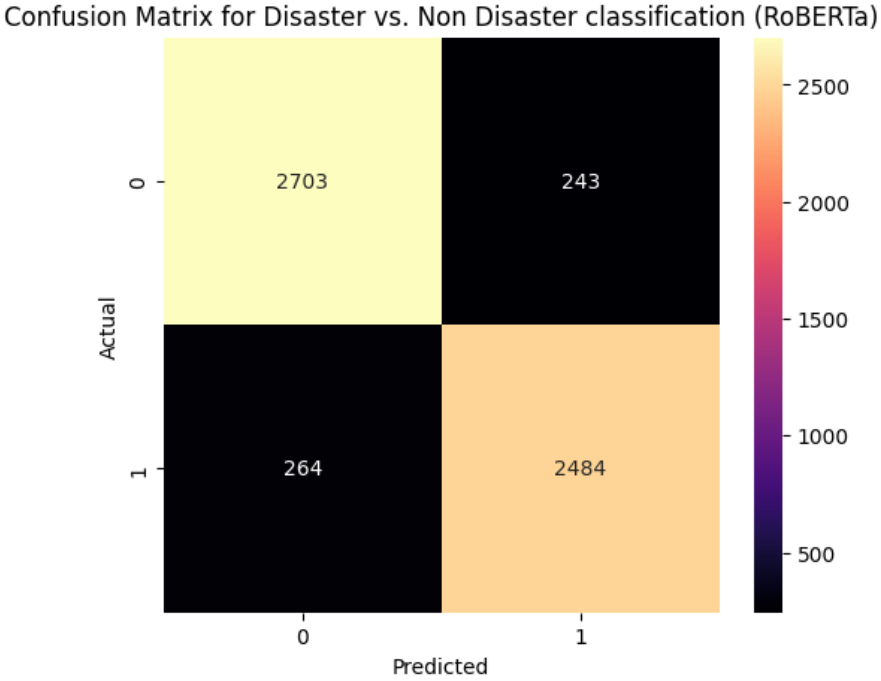

Figure 9: Disaster/Not Confusion Matrix (RoBERTa)

The confusion matrix of disaster/non-disaster classification is shown in **Figure 9**. The model appropriately recognizes 2703 non-disaster tweets and 2484 disaster tweets. There are 243 non-disaster tweets that are predicted to be disaster and 264 disaster tweets that are predicted to be non-disaster, which can be explained by ambiguous or context-bound textual information.

4.1.4.3 Image Model

The proposed system employs the same ResNet50 model as in the earlier step in the classification of disaster types. As the model architecture and training configuration is the same, its classification performance is cited based on **Figure 4**.

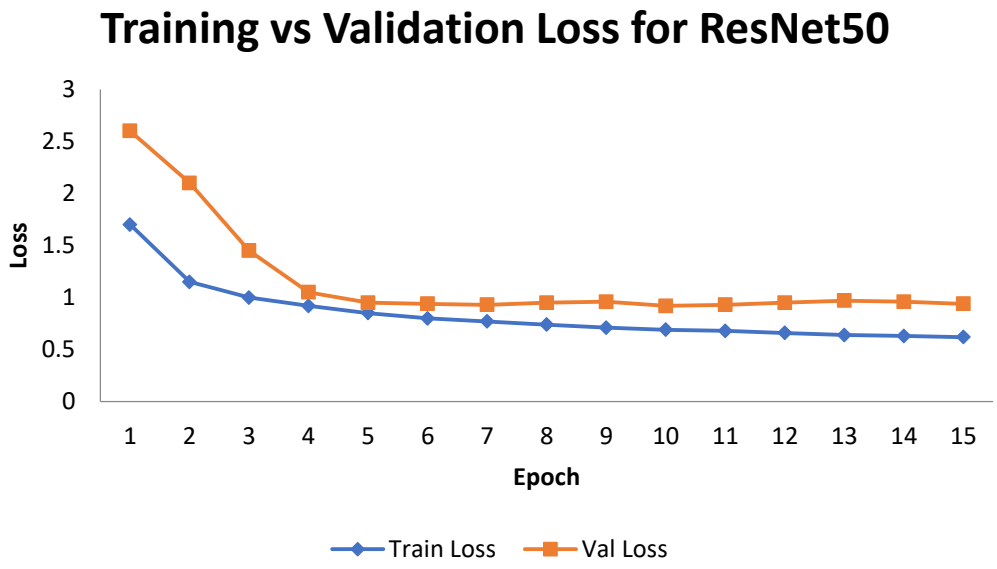

Figure 10: Training and validation loss curve for ResNet50 model

In addition, the training behavior of the ResNet50 model is illustrated in **Figure 10**, which shows the training and validation loss curves. The training loss gradually reduces with an approximate value of 1.7 to 0.6 which is an indicator of effective feature learning. The loss of validation also indicates a drastic drop during early epochs and then stabilizes to 0.9, indicating convergence of the model.

There is a narrow difference between training and validation loss, implying that the model can generalize well, although there are small changes in validation loss, which may imply insignificant changes in performance between epochs.

4.1.4.4 Final Decision-Level Fusion Model

The multitask RoBERTa model and the ResNet50 model are fused at the decision-level fusion to improve the overall system performance.

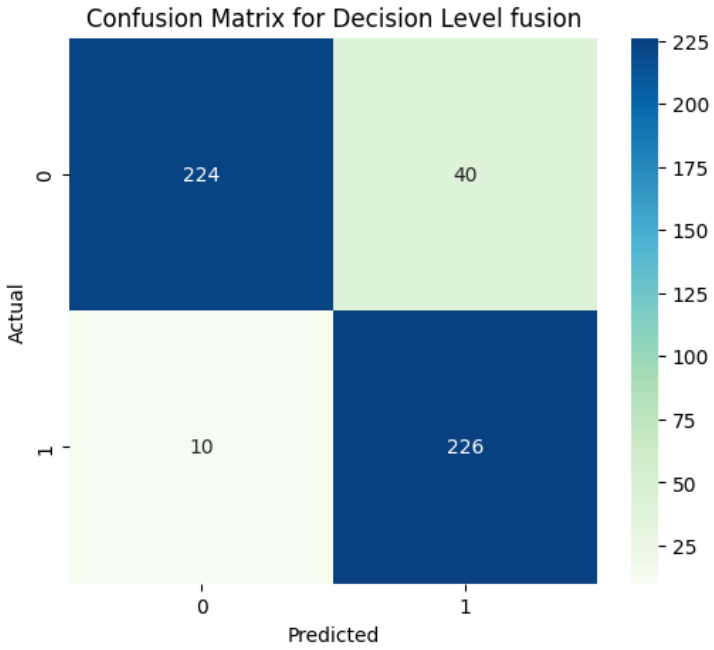

Figure 11: Final Fusion Confusion Matrix

The confusion matrix of the final fusion model is presented in **Figure 11**. Many correct predictions are made with 226 and 224 disaster and non-disaster instances respectively being correctly predicted. False positives and false negatives are comparatively low with 40 and 10 respectively.

This shows that the fusion method helps to combine both textual and visual data and makes classification more reliable and minimizes errors in comparison with separate models.

Overall, the suggested multitask multimodal system attains better performance, as it uses common textual representations, visual feature extraction, and decision-level fusion, which leads to a stronger and more effective disaster classification model.

4.1.4.5 Sample Predictions

To show how system words in real-world a deployable interface has been developed to try sample predictions, which are shown in **Figure 12**. This example show how the system processes input text and how it generates multiple outputs.

Figure 12(a,b) show how the system takes in text and gives out kinds of information. This includes what kind of disaster it is, if it is real what kind of incident it is, where it is and how urgent it is.

Figure 12(a): Sample real disaster-related tweet describing a boat accident

Figure 12(a) has a tweet about a boat accident in Mathura, Uttar Pradesh. The tweet says sorry about the accident. Gives some information about what happened. This is what real social media posts look like when they talk about disasters. They have feelings and facts all mixed together.

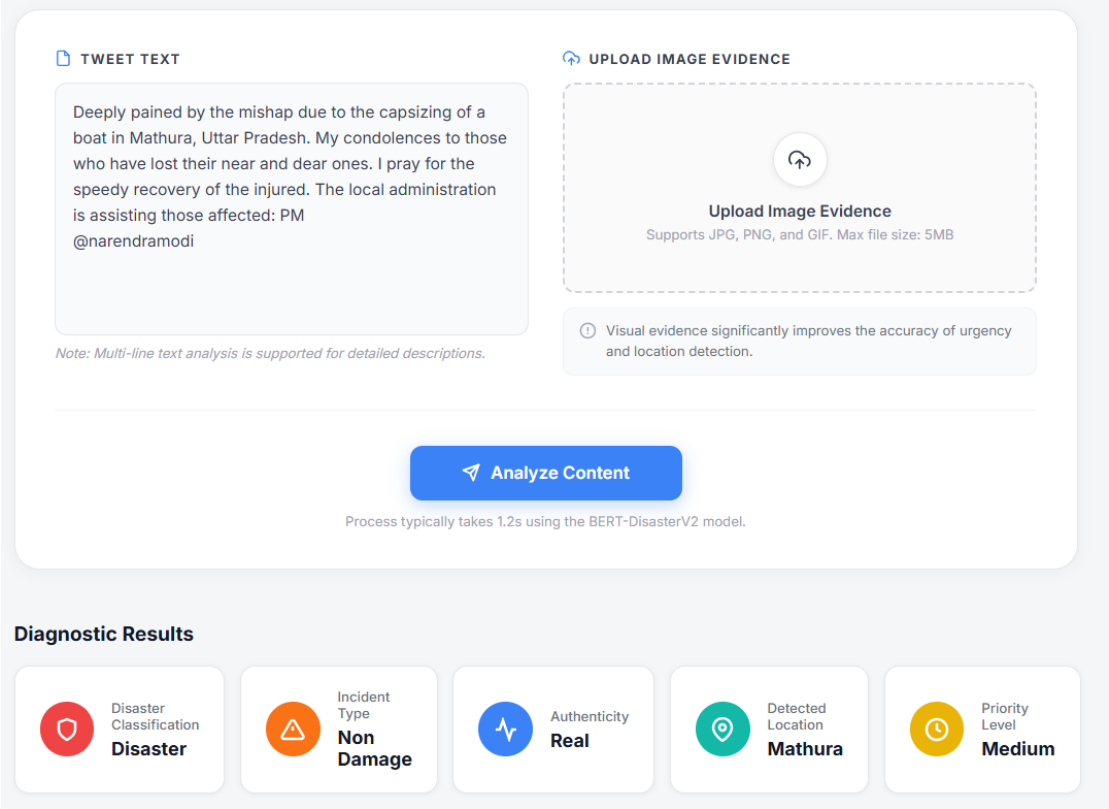

Figure 12(b): System output for real disaster tweet showing classification results

Figure 12(b) shows what the system says about the tweet. The system says it is a disaster because it has words like "mishap" and "capsizing" and it says where it happened. The system also says the tweet is real because it sounds like it is true. It makes sense.

The system does not think it is a disaster like a flood or a fire because the image was not provided to identify the type of disaster. The system can also find the place where it happened which's Mathura. The system says it is not very urgent because of how the tweet's written.

To check if the system works correctly another example has also been tried. To see if the system can tell when something is not true we tried another example.

Figure 13(a,b) shows that the system can tell when someone is trying to trick us with disaster news.

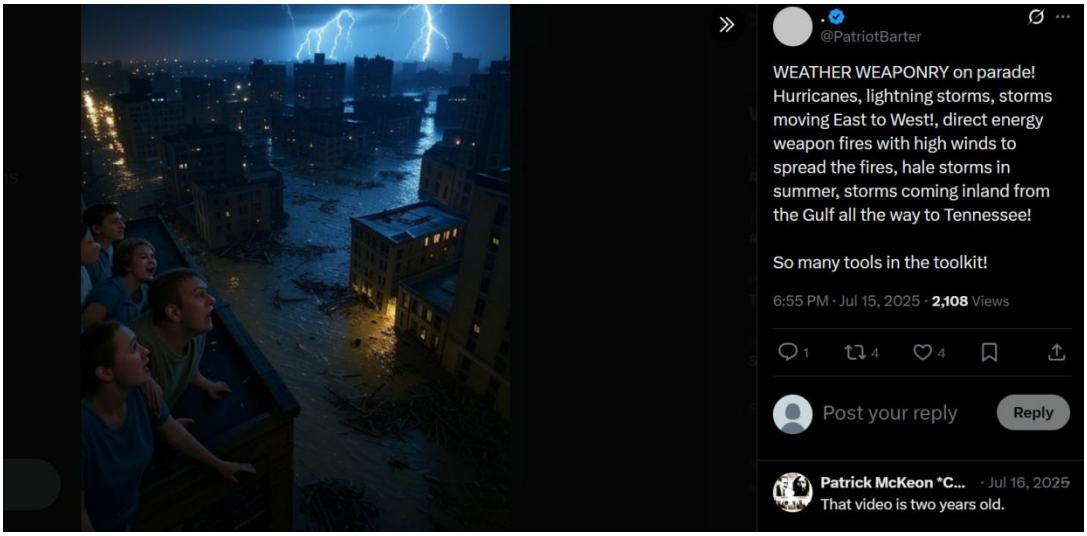

Figure 13(a): Sample social media post containing misleading disaster-related claims

Figure 13(a) has a post with a picture of a flooded city. Some writing that says some pretty crazy things. The writing says things that're not true and it uses language that is meant to get people excited.

Figure 13(b) shows what the system says about the post. The system says it is not a disaster because it does not have any facts about what happened. The system also says the post is fake because it is trying to trick us.

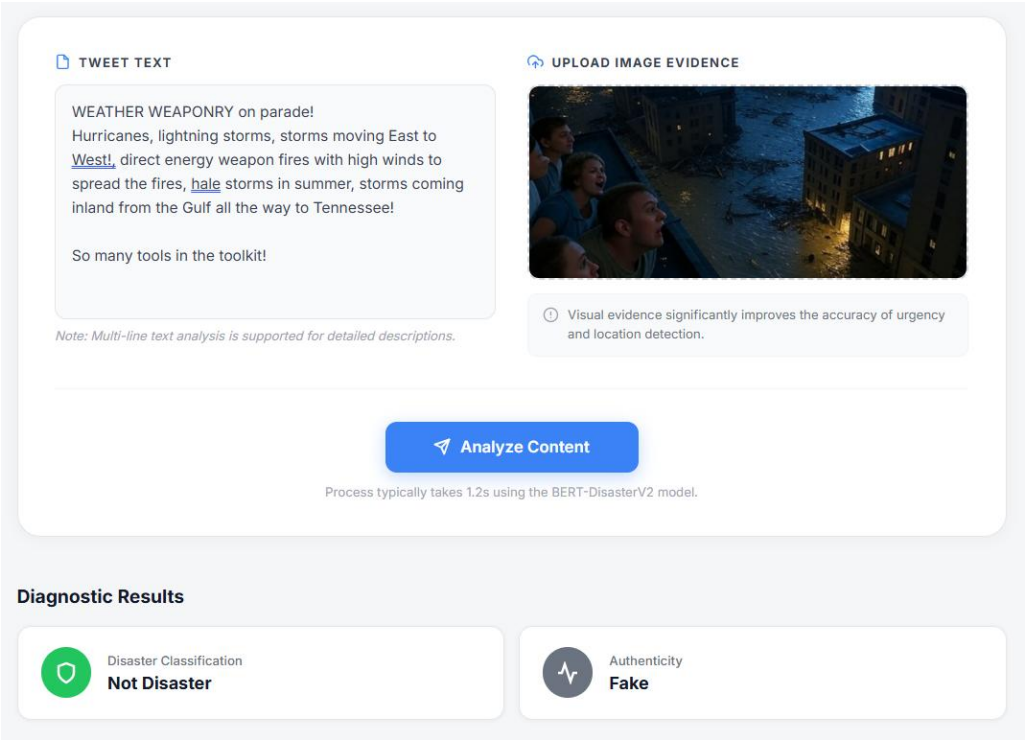

Figure 13(b): System output for fake content showing classification as non-disaster

The system says it is not a disaster because it does not have any details about what happened or where it happened. The post just says some things that are not true.

4.1.5 System Interface

4.1.5.1 Overview of the Dashboard

The system dashboard will be an interactive platform to monitor and analyze real-time disaster related information posted on the social media. It is mainly used to convert raw model predictions into meaningful and structured insights that may be used to assist in decision-making during critical scenarios as and when it is necessary.

The proposed system incorporates the outputs of several components into the dashboard, such as the text-based multitask classification model to determine the presence of a disaster and its authenticity (real/fake), the image-based classification model to detect the type of disaster, and the decision-level fusion mechanism to combine the two modalities. By bringing these outputs together into a single interface, the system ensures there is a global picture of disaster events and thus less ambiguity and enhanced interpretability.

The dashboard, as demonstrated in **Figure 14**, gives various integrated displays of the processed data. These consist of real-time intelligence feed of filtered disaster-based reports, a geospatial visualization of incidents and a summary of distribution of disaster types. The structured presentation allows users to recognize important events quickly, comprehend the context and examine their geographical distribution.

The dashboard will be a real-time disaster monitoring tool; only the information that is relevant and confirmed will be displayed after running it through the proposed models. This will make sure that users do not get bombarded with noisy or misleading information which is usually prevalent in social media feeds. Rather, the system emphasizes the high-confidence outputs, enhancing reliability and usability.

Moreover, the dashboard transforms multimodal predictions into actionable information by breaking down multimodal predictions into simple visual elements. Users are able to track the current incidents, determine their seriousness and have situational awareness without necessarily working with the underlying models. This renders the system applicable in the context of disaster management, emergency response, and social media monitoring platforms, where quick and informed decision-making is crucial.

All in all, the dashboard is the last interface of the suggested system, which links the high-level machine learning outputs to the real-life applicability and usefulness of the visualized and interactive dashboard.

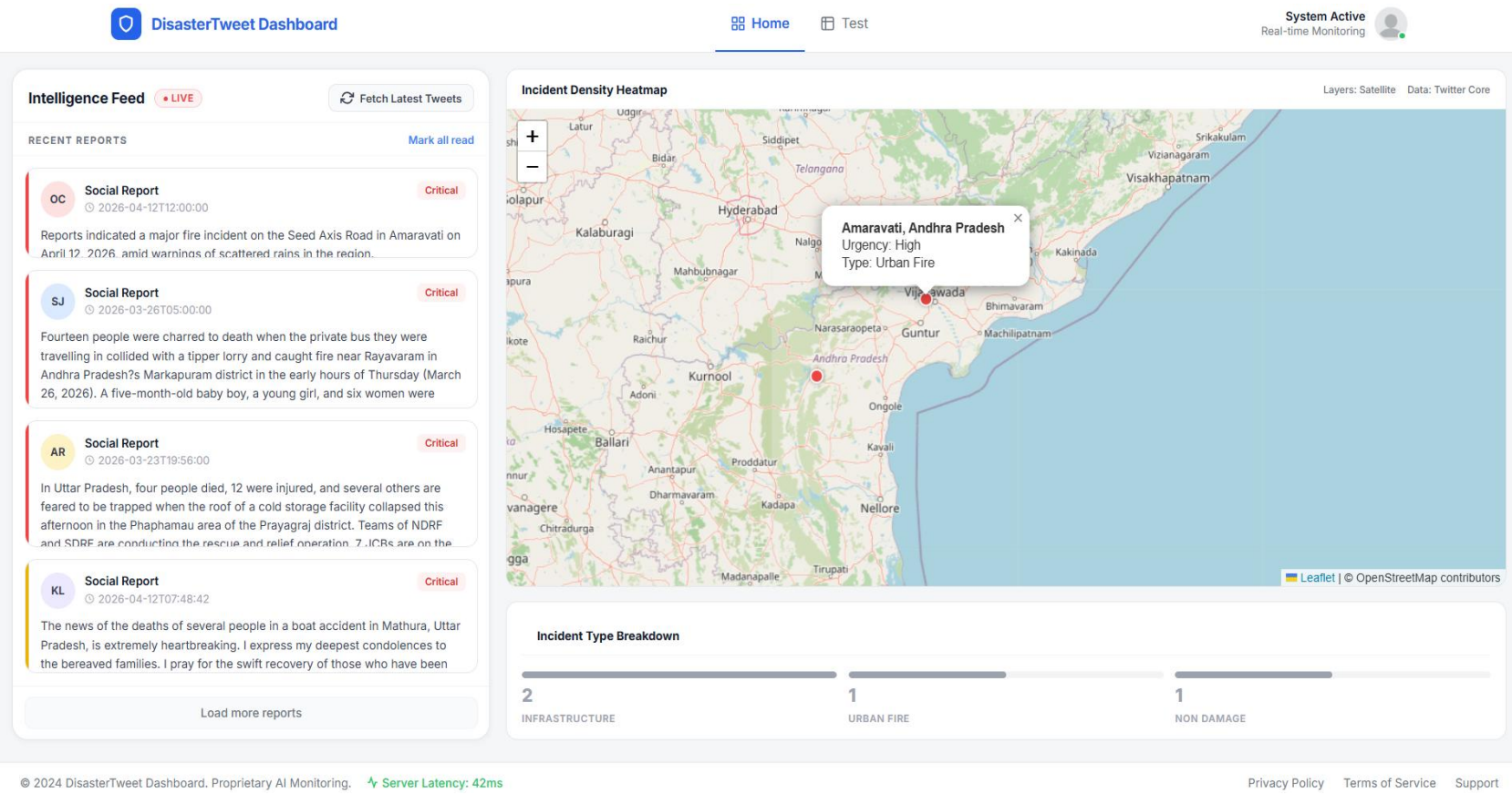

Figure 14: A Snapshot of the Disaster Tweet Dashboard

4.1.5.2 System Architecture Integration

The system is based on a hierarchical end-to-end pipeline that incorporates the data acquisition, multimodal processing, and visualization, so that real-time monitoring of disaster is possible. The social media media is used to gather the tweets and this is the main source of input of the system. Tweets are collected using an API, which is then used to start the workflow. The textual content or pictures found in these incoming tweets could be real-life information in the disaster related world.

After getting the data, a preprocessing phase is used to clean and standardize the inputs. This involves text normalization, tokenization, and simple filtering to eliminate noise, image resizing and transformation to prepare the data as an input to the model. This step achieves consistency and enhances effectiveness of further model predictions.

The textual data so processed is then subjected to a multitask RoBERTa model which, in the process, does two classification tasks: deciding whether the text is disaster-related and deciding whether the text is real or fake. This dual classification head representation can be easily used to extract contextual and semantic features of the text.

Simultaneously, when the image related to the tweet is provided, it is processed with the help of the ResNet50-based model. This model is in charge of categorizing the disaster type (flood, earthquake, or fire) in the image by extracting high-level visual features with deep convolutional layers.

In addition to the core classification, there are other modules that add to the contextual comprehension of the system. Location-related information found in the textual content is extracted with the help of Named Entity Recognition (NER). The resulting extracted location is then translated into geographical (latitude and longitude) co-ordinates with the aid of geocoding service, which facilitates spatial visualization. At the same time, sentiment analysis is conducted on the text to approximate the urgency or severity of the situation, which is an added value of information to make a decision.

The text and image model outputs are then fused together by a decision-level fusion mechanism. This integration step brings together the forecasts of both modalities to

enhance their overall accuracy and minimize uncertainty especially when either of the modalities might not be adequate.

After the fusion, the database only stores relevant and verified results, i.e., those that are disaster-related and real. The storage layer keeps well-organized data such as the content of tweets, the predicted labels, disaster type, urgency level, location coordinates, and timestamps, which allow the efficient retrieval and analysis of the data.

Lastly, the processed and stored data is displayed on the system dashboard wherein it is represented in the form of real-time feeds, geospatial maps, and statistical summaries. This is what makes up the pipeline, converting raw multimodal inputs into actionable insights that help in efficient monitoring and response to disaster events.

4.1.5.3 Functional Components of the Dashboard

The dashboard has been organized into various functional units that work together to convert model outputs into a decision-oriented and interpretable interface. Each of the components is created to perform a particular part of the data processing, visualization, and user interaction, so that the system can be informative and useful in its operations.

A) Real-Time Intelligence Feed

The real-time intelligence feed is the main source of information of the system and it shows dynamically retrieved reports of disasters on social media. The proposed models retrieve the Tweets using API and display them in this section.

At this step, a critical filtering process is used, and all tweets that meet this condition Disaster is true and Real is true are kept, are stored and exhibited. This will make sure only tested and valid disaster information is brought forward and all the noisy, misleading and irrelevant information that is so common in the social media streams is avoided.

Every item in the intelligence feed consists of organized content based on system pipeline, such as the content of the tweet, time, type of disaster expected, level of urgency, and location. Arranging this information in a brief and easy-to-read format,

the system allows users to evaluate the current incidents swiftly and prioritize urgent cases.

This component indicates how smart the system is because it not only identifies disasters but also authenticates the validity and adds contextual properties to the data.

B) Geospatial Visualization Module

The geospatial visualization module: This module is an extension that gives map-based visualization of the disaster events detected and allows spatial analysis of the incidents. In **Figure 15**, the disaster locations are plotted based on latitude and longitude measurements using the textual data as shown in the figure.

Location extraction is realized with the help of Named Entity Recognition (NER), which recognizes names of places in the text of the tweet. The resulting extracted locations are then translated into accurate geographical coordinates with the help of a geocoding service (Nominatim). The obtained coordinates are then plotted on the map as markers which depict the individual disaster events.

This module is important in improving situational awareness since it enables the users to visually analyze the distribution of disasters geographically. It assists in determining spatial patterns including clustering of incidents, concentration of certain type of disasters in a region, and high-risk areas. This information is also useful in disaster response planning, resource management, and real-time surveillance of the changing situation.

C) Incident Type Distribution

The incident type distribution component is a summary of the detected disaster types. It shows statistical data on the rates of occurrence of various types of disasters identified by the system, e.g., floods, fires, infrastructure damage, and others.

The component allows users to quickly grasp prevailing disaster trends in the data being monitored. Through the analysis of these distributions, trends can be determined like the growth of certain type of disaster in particular periods or areas. This underpins a higher level of analytical understanding and helps to make strategic decisions, as opposed to mere event-level monitoring.

D) User Interaction Module

The dashboard also has interactive capabilities to enable users to be active participants on the system. The Fetch Latest Tweets feature allows retrieving new data in real-time so that the dashboard could be updated with the latest information related to disasters.

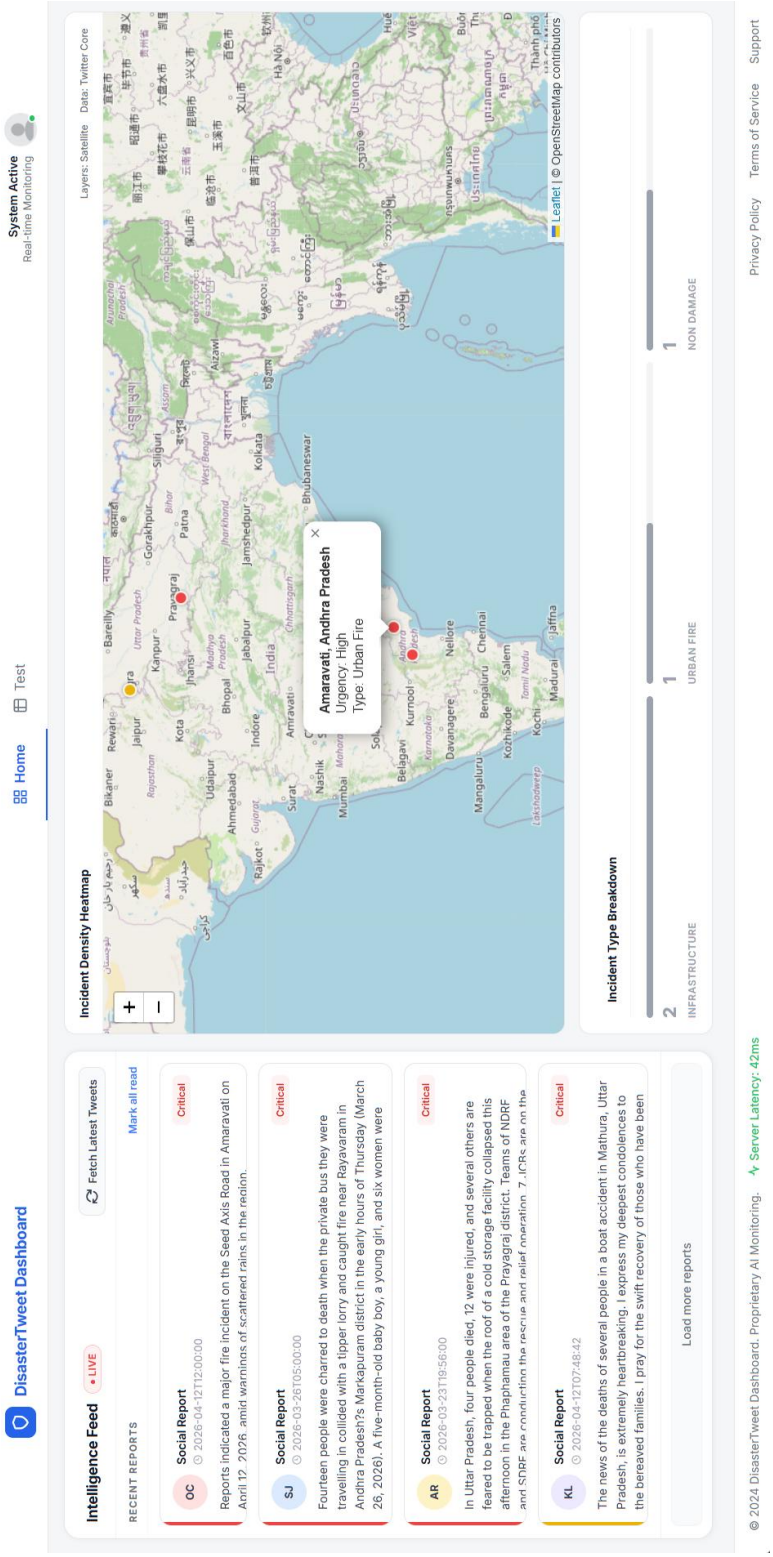

Figure 15: Interface snapshot showing real time tweets and Heatmap

Also, there is a separate testing interface to test system behavior manually. This enables the users to enter their own data and view model predictions, which helps to validate and verify system performance.

The interaction capabilities make the dashboard more usable, as the dashboard is not only automated in real-time monitoring but also allows manual control, which ensures the application in the real-world case.

4.1.5.4 Underlying Technologies

The system is developed with the aid of modern web and machine learning technologies, which will guarantee effective data processing, easy integration, and interactive visualization. React with Vite is used to create the frontend of the dashboard, which allows a dynamic user interface based on real-time data.

The backend is written in Flask (Python), which processes the data, makes inferences, and communicates among various components of the system. To store the data, MySQL is utilized to keep organized history of processed tweets, prediction outputs and related metadata.

To visualize geospatially, the system will use Leaflet with OpenStreetMap to enable the effective visualization of disaster information based on location on an interactive map.

Such combination of technologies makes sure that system is scalable, responsive and can process and visualize real-time multimodal data.

4.1.5.5 Data Processing and Storage Strategy

This system has a selective data processing and storage policy that only high-quality and relevant information can be stored to be analyzed and visualized. Once the multimodal pipeline has been traversed, each of the inputs is rated by the results of the classification models. Only the cases which are both disaster-related and valid (real) are accepted and then put into the database.

This filtering process is essential in removing noisy, misleading or irrelevant data in social media which is prevalent in real world data streams. The system keeps a clean and trustworthy dataset to downstream visualization and analysis by limiting storage to validated outputs.

The database schema will be structured to consist of all the necessary features needed to be properly monitored and interpreted. The stored records contain the text of the tweet, the picture (where possible), the predicted classification labels (disaster/not and real/fake), the type of disaster, the extracted location data, the geographical coordinates (latitude and longitude), the date when the event happened, and the approximate urgency level based on sentiment analysis.

This organized storage method facilitates effective querying, retrieval, and aggregation of information on disasters. It facilitates several functionalities of the dashboard, such as real-time feeds, geospatial mapping, and statistical summaries.

Such a selective storage system enhances data reliability and minimizes noise so that the system is only analyzing the information that can be acted upon and is reliable.

On the whole, the data processing and storage policy contributes to the strength of the system, as it helps to preserve the integrity of data and enables to manage information efficiently and at large scale.

4.1.5.6 System Workflow Summary

The entire system flow is based on a systematic pipeline that converts the raw social media information into understanding and actionable results. It starts with gathering of tweets posted by external sources via an API and this can contain both text and images.

The model processing stage is followed by the collected data passing through the textual content analysis with the multitask RoBERTa model as the textual content is analyzed to find out the disaster relevance and authenticity, and the image (where it is available) is analyzed with the ResNet50 model to identify the type of disaster. Simultaneously, these modules are supported, including Named Entity Recognition to extract location data, sentiment analysis to estimate urgency, and geocoding to transform extracted locations into geographical coordinates.

After the process of model processing a filtering step is used to only keep the inputs that are considered to be both disaster-related and authentic. This will make sure that only sound information and information that is relevant will continue to move along the pipeline.

The filtered results are subsequently stored in a structured database where all the relevant attributes such as predictions, location information, time and urgency are stored to be accessed and analyzed effectively.

Lastly, the data that has been stored is displayed via the dashboard whereby it is visualized through real-time feeds, geospatial maps, and aggregated statistics. This makes the pipeline complete by transforming the raw multimodal inputs to an intuitive and user friendly interface that facilitates good monitoring and decision making.

This simplified pipeline guarantees a smooth flow of data, elevated reliability of produced results, and the fluency of machine learning models and user-facing elements.

4.1.5.7 Practical Significance of Dashboard

The created dashboard can be seen as a valuable addition to practical applications as it simplifies the multimodal model outputs to make them applicable to real-world disaster monitoring and decision support. In contrast to the independent machine learning models, the dashboard allows observing and interpreting disaster-related phenomena in a work environment continuously.

Real-time monitoring of disasters is one of the main contributions of the system. Posting only verified and relevant information, the dashboard makes users follow the unfolding incidents by processing incoming data regularly and showing them on a social media. Situational awareness is also improved through the integration of geospatial visualization since the user can identify geographical locations where the incident has occurred as well as track the spread of the incident and identify groups of disaster occurrences.

Another critical aspect is misinformation control. In times of disaster, exaggerated, misleading or false information is common in social media. Such real/fake classification mechanism will ensure that only valid information is brought up to the surface and stored. This minimizes the chances of misinformation spread and assists in sustaining confidence in the system outputs, which is vital to both end-users and authorities.

The system also supports the emergency response and decision-making well. The dashboard provides an integrated picture of critical incidents by integrating information like disaster type, location, and urgency level. This enables the decision-makers to prioritize on responses, allocate resources in the most effective way and take timely action in high risk situations. Quick interpretation of multimodal data with no need to manually analyze the information saves a lot of time when responding to an emergency.

Also, the systematic and filtered delivery of the information reduces cognitive load among users to help them make a more informed decision with a reduced time to comprehend it. The dashboard therefore serves as an interface between state-of-the-art machine learning technology and the real-world implementation and deployment and therefore, the system is applicable to be used in disaster management systems, social media monitoring systems, and even public safety systems.

In general, the dashboard makes the proposed system more than a predictive model as it can real-time monitor the situation, enhance the reliability of information, and assist in the development of effective disaster response plans.

4.2 Result Analysis

This paragraph will provide a breakdown of the work of the suggested system by looking at the outcomes achieved at various levels of development, such as the original text-based system, the multimodal system and the final multitask system. The goal is to determine the contribution of each step towards the enhancement of the overall classification performance.

The confusion matrices and standard performance metrics are used to conduct the analysis and these metrics give insights on the accuracy of classification, error patterns, and model behavior. These models are compared to evaluate, systematically, the effectiveness of the inclusion of visual features and multitask learning in the process of optimizing the detection of disaster-related information.

4.2.1 Model-wise Performance Evaluation

The proposed system can be analyzed in terms of the individual contributions of text-based models, image-based models, and the fusion approach to assess the

performance of the proposed system. All components serve a specific purpose in the classification process.

(A) Text Model Performance

The text-based models have good capabilities of comprehending the contextual and semantic organization of tweets. The baseline DistilBERT model attains an accuracy of 0.85, whereas the RoBERTa model increases the accuracy to 0.87 in the multimodal stage and further to 0.8820 in the proposed system.

This advancement represents that RoBERTa is more competent in comprehending complicated linguistic structures, contextual relations, and domain-specific terminologies found within the social media information. Also, the real/ fake classification task has an accuracy of 0.8442, which indicates that the model was able to address subtle classification tasks.

Nevertheless, there are still issues related to the textual data. The wording of tweets can be ambiguous, sarcasm or lack of information or indirect references, thus misclassified. These considerations restrict the capability of text-only models to have ideal classification performance.

(B) Image Model Performance

ResNet50 model is applied to the classification of disaster type and its accuracy is 0.85, the balanced accuracy is 0.8509 and recall is 0.8554. This means that the model can be used to extract useful visual features and to tell the various disaster types apart.

Nonetheless, some restrictions are noted as a result of visual similarities between types of disasters. Categories like floods, landslides and infrastructure damage are sometimes overlapping with features like debris, water deposits and structural damage and it is hard to distinguish between the two categories in the model.

In general, the image model works well in well-defined visual context, but its performance is influenced in cases where the visual patterns cannot be clearly separated.

(C) Fusion Model Performance

Fusion model shows the optimal performance of all approaches. The fusion model in the multimodal stage has an accuracy of 0.91 and in the proposed system, it has an accuracy of 0.9050 with an F1-score of 0.9097.

The complementary character of textual and visual data can be attributed to the improvement in performance. Whereas the textual models are used to acquire meaning in context, image models are used to present visual evidence. The system can eliminate ambiguities in each model by integrating both modalities.

As an example, visual cues can be used to rectify misleading textual information and textual context can be used to interpret ambiguous images. The synergy leads to a higher classification rate, lower misclassification, and a better system robustness.

Therefore, it is the fusion model that is important in the realization of effective and congruent performance in real world conditions.

4.2.2 Error and Misclassification Analysis

Although high performance is attained, some trends of misclassifying are found in all the models. Such errors are mainly caused by the complexity and ambiguity of real-world data.

The existence of ambiguous or indirect language is one of the key causes of error in text-based models. Incomplete information, figurative language or exaggerated statements are common in tweets and may cause false positives where non-disaster content is identified as disaster-related. On the same note, tweets that are associated with a disaster, but do not contain specific keywords or context can be categorized as non-disaster, and this will lead to false negative.

Subtle misinformation and partially true statements can be a challenge in the case of real/fake classification. The model might not be able to differentiate authentic and misleading content in situations where the content uses similar patterns of language.

In the case of image-based classification, misclassification is caused by overlapping visual characteristics in various categories of disasters. Indicatively, floods, landslides and infrastructural damage tend to have similar visual features like debris, water, and

damaged buildings. This creates misunderstanding of classes particularly when the distinctive features are not obvious.

The absence of contextual information in individual modalities is another factor of contribution. A text and pictures alone might not be enough to describe the situation and offer enough semantic meaning respectively. The fusion model to a great extent takes care of this limitation, minimizing the errors by integrating the two sources of information.

The misclassification patterns outlined overall indicate the difficulty with managing noisy, unstructured, and multimodal real-world data, as well as the role of multiple modalities in enhancing performance.

4.2.3 Quantitative Performance Comparison

In order to present a thorough assessment of the system, the results of various models are compared according to conventional measures, including accuracy, precision, recall, and F1-score. Based on the **Table 2**, it is clear that RoBERTa is superior to DistilBERT in text-based classification tasks, and improved contextual representation is beneficial. ResNet50 model performs consistently in image-based classification, which implies that it is effective in creating features.

Table 2: Comparison of Performance metrics of different models used

Model	Task	Accuracy	Precision	Recall	F1-score
DistilBERT	Disaster Text (Baseline)	0.85	0.8501	0.8572	0.8536
RoBERTa	Disaster Text (Multimodal Stage)	0.87	0.877	0.872	0.873
ResNet50	Disaster Image	0.85	0.8509	0.8554	0.8531
Fusion	Text + Image (Multimodal)	0.91	0.908	0.903	0.905
RoBERTa	Disaster Text (Proposed)	0.8820	0.8873	0.8820	0.8836
RoBERTa	Real/Fake Text (Proposed)	0.8442	0.8509	0.8554	0.8467
Fusion	Text+Image (Proposed System)	0.9050	0.898	0.903	0.9097

The highest performance is achieved by the fusion models, both in the multimodal stage and the proposed system. The fusion strategy is always able to increase the accuracy and F1-score because it uses complementary information between text and images.

Even though the real/fake classification task results in slightly lesser accuracy than disaster detection, the accuracy remains high, which indicates the ability of the multitask model to deal with multiple goals at the same time.

Generally, the quantitative comparison illustrates that the multimodal fusion and multitask learning are very important in improving the performance of the system as opposed to the single models.

4.3 Observations from the Work

This section focuses on the major observations that were obtained by the experimental analysis of the suggested multimodal fake news detector of disaster news.

1. Multimodal Learning Enhances Decision Reliability

A combination of textual and visual information contributes greatly to the consistency and reliability of predictions. Although individual models have some constraints, the integrated model minimizes uncertainty in classification decision making. This implies that multimodal learning is more apt in dealing with complex and noisy social media data as opposed to single-modality approaches.

2. Textual Features Provide Strong Discriminative Power

The text-based model has always shown good performance over the image-based model. This implies that textual information is more precise and explicit in order to know the relevance and authenticity of disasters. Contextual language models are especially useful in the capture of semantic meaning that is very important in classification tasks using social media data.

3. Visual Information Acts as Supporting Evidence

In spite of the relatively poor standalone performance, image model plays a significant role when integrated with textual data. Visual characteristics can serve as auxiliary information that assists to confirm or narrow down predictions particularly

where the textual information is ambiguous or deceptive. This underscores the complementary nature of visual data in the multimodal systems.

4. Fusion Mechanism Reduces Ambiguity

The decision-level fusion technique efficiently reduces misclassification, capitalizing on the advantages of both modalities. It assists in solving conflicts when one of the modalities makes uncertain or incorrect predictions. This shows that fusion is necessary to enhance robustness and balance performance in various situations.

5. Impact of Data Complexity on Model Performance

Ambiguous language, incomplete information, and a visual resemblance of disaster categories provide the challenge of classification. These elements have the effect of both false positives and false negatives, meaning that real-world data in social media is always difficult to model accurately.

6. Improved Generalization through Diverse Data

The fact that several datasets are used also helps to better generalize the model. The ability to expose the system to diverse data distributions allows it to be effective to the different disaster-related content types, minimizing overfitting risks and enhancing adaptability.

7. Practical Applicability of the Proposed System

The system has a high potential of actual implementation in real-world scenarios like the disaster monitoring system, misinformation detector and emergency response system. The fact that it can process text and visual input increases its functionality in dynamic and informative settings.

Conclusion

The observations suggest that the proposed multimodal framework is a powerful and scaled solution to detect fake disaster news. The textual knowledge, analysis and fusion techniques help in boosting performance and reliability of the system to a very great extent.

CHAPTER 5

Conclusion and Future Work

5.1 Conclusion

This project introduces a multimodal deep learning system of identifying fake news that are related to disasters on social media using a combination of both text and visual data. The system is developed to a text-based base model then to a multimodal architecture and finally a multitask framework, showing a progressive increase in performance.

The textual classification is performed with a fine-tuned RoBERTa model, whereas a model based on ResNet50 is utilized to draw visual features of images related to disasters. The results of the two models are then fused together by the use of a decision-level fusion method that greatly improves the performance in classification because of the complementary information in the two modalities.

The experimental findings indicate that multimodal and multitask method is superior to unimodal methods, and it has better accuracy, a decreased misclassification rate, and ability to deal with ambiguous data. The fusion mechanism is important in enhancing reliability by addressing the discrepancies between text and image predictions.

On the whole, the suggested system is a successful solution to the misinformation detection related to disaster and has good perspectives of practical implementation in real-life scenarios in social media monitoring, disaster management, and community safety systems.

5.2 Future Study

Though the suggested system shows good performance, a number of directions could be pursued to increase its capabilities:

- **Incorporation of Additional Modalities**: It is possible to incorporate video and audio data to expand the system in the future. Video-based temporal information can give more contextual information and enhance the detection accuracy.

- **Advanced Fusion Techniques:** Alternatively to decision-level fusion, feature-level fusion or attention-based fusion can be considered to more effectively represent interactions between text and visual features.
- **Real-Time Deployment:** The existing system is based on fixed datasets. The next step of work can be to concentrate on real-time deployment based on the live social media streams (e.g., Twitter APIs) to perform continuous monitoring and detect misinformation at an early stage.
- **Dataset Expansion and Multilingual Support:** Large, diverse, and multilingual datasets can be trained to enhance generalization and allow the system to be used in other regions and languages.
- **Explainable AI Integration:** Visualizing attention or saliency maps can enhance the transparency and trust in model predictions particularly in critical applications.

These future directions will be focused on enhancing the scalability, flexibility and relevance of the proposed system in the real world.

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APPENDIX

Some screenshots of the code snippet:

```python
# ================= TEXT CLEANING =================

def clean_text(text):
    text = str(text)

    # Lowercase
    text = text.lower()

    # Remove URLs
    text = re.sub(r"http\S+|www\S+", "", text)

    # Convert emojis to text
    text = emoji.demojize(text, delimiters=(" ", " "))

    # Replace hashtags: #flood → hashtag_flood
    text = re.sub(r"#(\w+)", r"hashtag_\1", text)

    # Remove mentions
    text = re.sub(r"@\w+", "", text)

    # Remove extra symbols
    text = re.sub(r"[^a-zA-Z0-9_\s]", "", text)

    # Remove extra spaces
    text = re.sub(r"\s+", " ", text).strip()

    return text

df["text"] = df["text"].apply(clean_text)
```

Figure 16: Data preprocessing of textual data

```python
# ================= TOKENIZER =================
tokenizer = RobertaTokenizer.from_pretrained("roberta-base")
MAX_LEN = 128
```

```
'(ProtocolError('Connection aborted.', ConnectionResetError(10054, 'An existing connection was forcibly closed by the remote
Retrying in 1s [Retry 1/5].
```

```python
# ================= DATASET =================
class MultiTaskDataset(Dataset):
    def __init__(self, df):
        self.texts = df["text"].tolist()
        self.disaster = df["disaster"].tolist()
        self.real = df["real"].tolist()

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        encoding = tokenizer(
            self.texts[idx],
            padding="max_length",
            truncation=True,
            max_length=MAX_LEN,
            return_tensors="pt"
        )

        return {
            "input_ids": encoding["input_ids"].squeeze(0),
            "attention_mask": encoding["attention_mask"].squeeze(0),
            "disaster": torch.tensor(self.disaster[idx]),
            "real": torch.tensor(self.real[idx])
        }

train_loader = DataLoader(MultiTaskDataset(train_df), batch_size=16, shuffle=True)
val_loader = DataLoader(MultiTaskDataset(val_df), batch_size=16)
```

Figure 17: Tokenization of input text using RoBERTa tokenizer

```python
# ================= MODEL =================
class MultiTaskRoBERTa(nn.Module):
    def __init__(self):
        super().__init__()
        self.roberta = RobertaModel.from_pretrained("roberta-base")
        hidden = self.roberta.config.hidden_size

        self.dropout = nn.Dropout(0.3)
        self.disaster_head = nn.Linear(hidden, 2)
        self.real_head = nn.Linear(hidden, 2)

    def forward(self, input_ids, attention_mask):
        out = self.roberta(input_ids=input_ids, attention_mask=attention_mask)
        cls = out.last_hidden_state[:, 0, :]
        x = self.dropout(cls)

        return self.disaster_head(x), self.real_head(x)

model = MultiTaskRoBERTa().to(device)
optimizer = AdamW(model.parameters(), lr=2e-5)
```

Figure 18: Multitask RoBERTa architecture with dual classification heads

```python
for batch in pbar:
    optimizer.zero_grad()

    input_ids = batch["input_ids"].to(device)
    mask = batch["attention_mask"].to(device)
    d_labels = batch["disaster"].to(device)
    r_labels = batch["real"].to(device)

    out_d, out_r = model(input_ids, mask)

    loss = 0

    # disaster
    mask_d = d_labels != -1
    if mask_d.sum() > 0:
        loss_d = disaster_criterion(out_d[mask_d], d_labels[mask_d])
        loss += loss_d
        total_loss_d += loss_d.item()

        preds_d.extend(torch.argmax(out_d[mask_d], 1).cpu().numpy())
        labels_d.extend(d_labels[mask_d].cpu().numpy())

    # real/fake
    mask_r = r_labels != -1
    if mask_r.sum() > 0:
        loss_r = real_criterion(out_r[mask_r], r_labels[mask_r])
        loss += loss_r
        total_loss_r += loss_r.item()

        preds_r.extend(torch.argmax(out_r[mask_r], 1).cpu().numpy())
        labels_r.extend(r_labels[mask_r].cpu().numpy())

    loss.backward()
    optimizer.step()
```

Figure 19: Fine-tuning of RoBERTa base with textual dataset

```python
# model finetune

base_model = ResNet50(
    weights="imagenet",
    include_top=False,
    input_shape=(224,224,3)
)

# TRUE fine-tuning
base_model.trainable = True

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(256, activation="relu")(x)
x = Dropout(0.4)(x)
outputs = Dense(train_gen.num_classes, activation="softmax")(x)

model = Model(inputs=base_model.input, outputs=outputs)

model.compile(
    optimizer=Adam(learning_rate=LR),
    loss=CategoricalCrossentropy(label_smoothing=0.1),
    metrics=["accuracy"]
)

model.summary()
```

Figure 20: Fine-tuning of ResNet50

```python
# training and saving best model

checkpoint = ModelCheckpoint(
    "best_resnet50_model.keras",
    monitor="val_loss",
    save_best_only=True,
    verbose=1
)

early_stop = EarlyStopping(
    monitor="val_loss",
    patience=5,
    restore_best_weights=True
)

history = model.fit(
    train_gen,
    validation_data=val_gen,
    epochs=EPOCHS,
    class_weight=class_weight,
    callbacks=[checkpoint, early_stop]
)
```

Figure 21: Training ResNet50 model

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